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What Counts as AI Sycophancy? A Taxonomy and Expert Survey of a Fragmented Construct

Meryl Ye[1], Lujain Ibrahim[2], Jessica Y. Bo[3], Myra Cheng[4], Ida Mattsson[1], Daniel Vennemeyer[5], Robert Kraut[1], Steve Rathje[1, 6]

[1]Carnegie Mellon University
[2]University of Oxford
[3]University of Toronto
[4]Stanford University
[5]University of Cincinnati
[6]New York University
merylye@cmu.edu

Abstract

AI sycophancy has become a prominent concern in large language model (LLM) research. Yet the term lacks a consistent definition and has been applied to behaviors ranging from agreeing with a user's false claim to excessively praising the user to withholding corrective feedback. When researchers, companies, and policymakers use the same term to describe different behaviors, evaluation results become difficult to compare, mitigation strategies fail to transfer, and systems that are resistant to one form of sycophancy continue exhibiting other forms. To address this, we make two contributions. First, we reviewed 70 papers on AI sycophancy to develop a taxonomy of how the behavior has been defined and measured. The taxonomy distinguishes (1) whether a model is sycophantic toward a user's positions and beliefs, or toward the user's broader personal traits and emotions, and (2) whether this occurs through explicit, direct language or more implicit, subtle behaviors such as framing, omission, or tone. Mapping existing literature to our taxonomy reveals that current research has focused on overt forms of sycophancy toward users' beliefs, leaving more subtle and person-directed behaviors relatively understudied. Second, we surveyed 106 experts in AI sycophancy and related fields to examine whether researchers agree on which model behaviors are sycophantic. While experts are nearly unanimous in believing that sycophancy is a significant problem in current AI systems (94.3% agree), they disagree substantially on which specific behaviors qualify. Together, these findings demonstrate that AI sycophancy is a broad family of behaviors with different measurement challenges, intervention requirements, and governance implications. Our taxonomy provides a shared vocabulary for understanding and addressing these behaviors. We discuss ways in which the taxonomy can inform evaluation design as well as both technical and regulatory interventions.

Introduction

Sycophancy has become a prominent concern in large language model (LLM) research. The topic has accumulated a growing body of empirical findings, evaluation benchmarks, and proposed mitigation strategies (Perez et al. 2023; Sharma et al. 2024; Wei et al. 2023). Academic literature on the topic has expanded rapidly in recent years, and public attention has risen accordingly (see Figure 1). Recent empirical findings show that sycophantic interactions increase attitude extremity and overconfidence (Rathje et al. 2025), impair task performance and learning (Bo et al. 2025), and alter users' social judgments and prosocial behavior (Cheng et al. 2026a; Ibrahim et al. 2026). Yet the term lacks a consistent definition. The range of behaviors it has been used to describe reveals a *fragmented construct* that is rapidly expanding without a shared conceptual anchor.

Construct fragmentation occurs when different studies use the same label for behaviors that differ in form, mechanism, and measurement (Flake and Fried 2020). While breadth in a construct is not inherently problematic, issues emerge when that breadth leads to findings that cannot be meaningfully compared or aggregated (Anvari et al. 2025). This fragmentation is evident in how recent studies operationalize sycophancy. Although these studies claim to examine the same phenomenon, they operationalize sycophancy as different behaviors in different settings. Yet findings from those studies are often treated interchangeably to benchmark models, evaluate mitigation strategies, or support safety claims. As a result, benchmarks built to detect sycophancy as factual capitulation cannot detect sycophancy as implicit framing bias. Mitigation strategies effective against agreement with users may not address sycophancy when it is expressed as praise of the user's character or abilities, and model specifications anchored to one set of behaviors leave others entirely unaddressed. Figure 1 provides an overview of these conceptual expansions over time.

These inconsistencies extend beyond academic studies into model evaluation and deployment. With the launch of GPT-5 in August 2025, OpenAI reported that its model incorporated changes designed to reduce sycophancy (Singh et al. 2025). However, evaluations on the ELEPHANT (Evaluation of LLMs as Excessive SycoPHANTs) benchmark (Cheng et al. 2026b), which examines social validation, indirectness, and framing in open-ended advice contexts, show that GPT-5 remains substantially sycophantic in the open-ended advice and interpersonal contexts that the benchmark probes. Similar discrepancies appear across benchmarks. SycEval (Fanous et al. 2025), which opera-

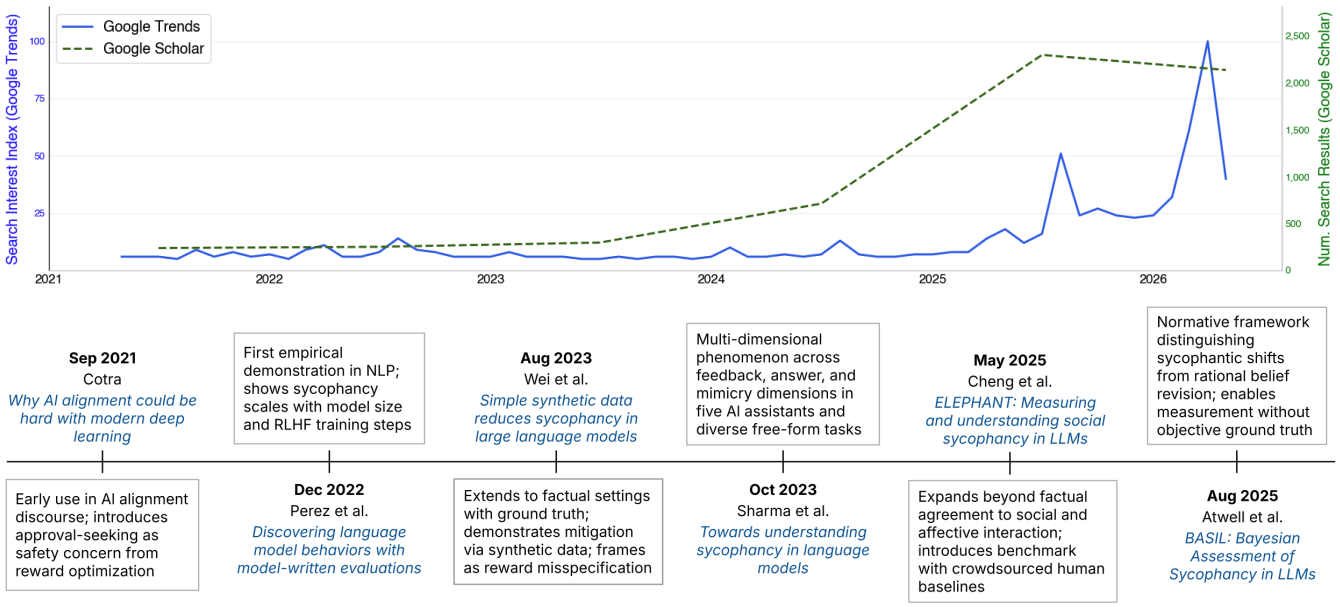


Figure 1: *Top*: Proxy measures of public and academic attention to the term "sycophancy" over time. Academic interest began rising in 2024, preceding the largest spikes in public search interest observed during 2025–2026 (Google 2026). *Bottom*: Overview of key conceptual expansions in AI sycophancy research. This timeline is not exhaustive.

tionalizes sycophancy as susceptibility to factual rebuttals, ranks Gemini as the most sycophantic model tested. In contrast, ELEPHANT ranks Gemini as the least sycophantic. Each benchmark operationalizes a different set of behaviors, and treating them as equivalent assessments of the same construct makes direct comparison difficult to interpret.

To address these challenges, this paper makes two contributions. First, through a review of 70 papers on AI sycophancy, published from 2023 to 2026, we examine how the construct has expanded over time, applied to increasingly diverse settings, tasks, and behavioral assumptions. This historical analysis reveals a rapidly growing but uneven research landscape, where new definitions, benchmarks, and mitigation strategies have proliferated without a shared definitional framework. Building on these patterns, we introduce a descriptive taxonomy of AI sycophancy organized along two orthogonal dimensions: *Referent*—whether the model responds to the user's expressed positions and claims (Position) or to the user as a person (Person)—and *Explicitness*—whether the behavior is expressed overtly or through subtler means such as framing, omission, or tone. Within each referent, we distinguish sub-referents: Verifiable versus Subjective for Position behaviors, and Traits versus Emotions for Person behaviors. This taxonomy provides a shared vocabulary for comparing behaviors that have previously been studied under the same label, but operationalized in different ways. Our analysis further reveals a strong concentration of prior work on explicit forms of sycophancy tied to users' verifiable claims, while subtler forms expressed through framing, omission, or behaviors targeting users as persons receive comparatively less empirical attention.

Second, we report results from a survey of 106 experts in AI sycophancy and related domains designed to examine whether researchers apply the label consistently when judging concrete model behaviors. We find that experts are nearly unanimous in agreeing that sycophancy is a significant problem in current AI systems ($M = 6.21$, $SD = 0.91$ on a 7-point Likert Scale; 94.3% agree), yet show substantial variation in which specific behaviors they consider sycophantic. Experts broadly agree that behaviors directed at users' positions are sycophantic whether expressed explicitly or through subtler means such as selective framing. However, for behaviors directed at the user as a person, experts are mixed: only explicit forms such as flattery and unwarranted praise are reliably recognized as sycophantic, while behaviors that convey the same deference through tone, softened feedback, or avoidance of critique are not.

These findings suggest that AI sycophancy comprises a family of behaviors whose boundaries remain actively contested. Expert disagreements reflect differences in where researchers draw those boundaries and potentially how they interpret the consequences of sycophantic behavior across sustained interaction. As researchers, AI companies, and policymakers increasingly invoke the term, progress will depend on clearly specifying which types of behavior are being measured, mitigated, and regulated. Our taxonomy and expert survey provide a foundation for these specifications, and we discuss their implications for evaluation methods, mitigation strategies, and corporate and legislative governance.

Sycophancy as a Fragmented Construct

Divergent Definitions

Across major English dictionary definitions, the term *sycophancy* and its variants (e.g., sycophant, sycophantic) cluster around a consistent set of themes: insincere praise directed at those with power or status, motivated by self-interest (Rehani et al. 2026). Early AI safety discourse often preserved this agentive framing, extending it into accounts of sycophancy as strategic or deceptive behavior (Cotra 2021; Denison et al. 2024). However, recent AI sycophancy research has largely set aside these motivational assumptions, treating sycophancy as an approval-seeking behavioral pattern that arises from training on human feedback. Throughout this paper, we follow the convention of treating sycophancy as a pattern in model outputs without assuming intentional or strategic motivations, since intent is not directly observable in most empirical settings. The construct has nonetheless been defined in increasingly divergent ways.

Although early AI research used the term in relatively consistent ways, later work has operationalized sycophancy in increasingly varied ways, differing in settings, tasks, and behavioral assumptions. Initial empirical definitions focused on agreement with user-stated positions in structured, proposition-based tasks where models could be evaluated against the "ground truth" (i.e., a preferred or externally specified response) (Perez et al. 2023; Sharma et al. 2024). Subsequent work sought to explain the training-level causes of sycophancy, attributing sycophancy to reward misspecification (models trained on human approval signals learn to generate outputs that elicit favorable feedback rather than accurate information) and showing it can be reduced through synthetic-data fine-tuning Wei et al. (2023). Denison et al. (2024) extend this account more broadly by situating sycophancy within specification gaming (the tendency of models to satisfy the literal terms of a training objective while violating its intent), showing that models trained to seek user approval in simple interactions can generalize to more strategic reward-seeking behaviors and that standard harmlessness training does not reliably prevent this shift.

A limitation of these early accounts of sycophancy is that they cannot be applied in settings where no ground truth exists. Atwell et al. (2025) address this by redefining sycophancy as the extent to which a model shifts toward a user's stated position more than a Bayesian-rational agent would after observing the same evidence. This framework enables measurement in subjective and uncertain settings, but introduces new questions about what determines a rational baseline.

Parallel work expands the construct into interpersonal and affective domains where ground truth similarly does not exist. Cheng et al. (2026b) define sycophancy as excessive face-preservation (i.e., the tendency to affirm users' self-image through validation, indirectness, framing acceptance, and moral endorsement) and introduce ELEPHANT, a benchmark for measuring these behaviors against crowdsourced human baselines. A related finding is that training LLMs to be warm and empathetic makes them substantially more sycophantic (Ibrahim, Hafner, and Rocher 2026), raising the question of where appropriate affective responsiveness ends and sycophancy begins.

Further work illustrates the construct's heterogeneity from multiple angles. Vennemeyer et al. (2025) show that sycophantic agreement and sycophantic praise are functionally separable in model representations and can be independently steered, suggesting that even seemingly similar approval-seeking behaviors are distinct. Rathje et al. (2025) experimentally disentangle validating language from the one-sided provision of factual information, showing that these components produce different downstream effects: validation primarily increases enjoyment and positive perceptions of the AI chatbot, while one-sided factual statements primarily increase attitude extremity. Rehani et al. (2026) find that perceived sycophancy decomposes into three factors (Uncritical Agreement, Obsequiousness, and Excitement) which differ in how negatively users perceive them. Across these studies, behaviors commonly labeled as sycophancy differ not only in surface form but also in internal representation, downstream effects, and how they are perceived by users.

Why Fragmentation is a Problem

As the literature has expanded, researchers have introduced increasingly specialized subtypes of sycophantic behaviors. Sharma et al. (2024) propose answer sycophancy (changing answers to align with user beliefs), feedback sycophancy (changing feedback to align with user preferences), and mimicry sycophancy (repeating user errors or stylistic patterns). Fanous et al. (2025) distinguish progressive sycophancy (agreeing with incorrect user rebuttals) from regressive sycophancy (revising correct answers after user rebuttal). Cheng et al. (2026b) introduce four dimensions of social sycophancy: validation, indirectness, framing, and moral endorsement. Du et al. (2025) propose informational, cognitive, and affective sycophancy. These distinctions are not mutually exclusive and do not neatly map onto one another, resulting in a patchwork of labels without shared criteria for what counts as sycophancy.

This fragmentation is directly reflected in evaluation frameworks such as ELEPHANT (Cheng et al. 2026b), VISE (Video-LLM Sycophancy Benchmarking and Evaluation) (Zhou et al. 2025), BASIL (Bayesian Assessment of Sycophancy in LLMs) (Atwell et al. 2025), and SycEval (Fanous et al. 2025), which operationalize different forms of sycophantic behavior without situating them within a shared construct space. As a result, the apparent body of evidence rests on observations from benchmarks that measure different behaviors, making cumulative claims difficult to sustain. The absence of shared construct boundaries also creates openings for broader conceptual reinterpretation. For instance, Shi et al. (2026) absorb sycophancy into a unified taxonomy of LLM deception, treating it as a behavioral subtype because sycophantic responses may lead users to form false beliefs. Where early safety discourse assumed strategic intent, this reinterpretation extends the construct from observable behavioral outputs to their downstream epistemic effects, which most empirical studies of sycophancy do not directly measure.

These disagreements reflect differences in the conceptual boundaries researchers place around the term. When studies use the same label for behaviors that differ in form, setting, measurement strategy, and likely mechanism, it becomes difficult to compare findings, determine whether mitigation strategies transfer, or assess industry commitments to reducing sycophancy. These behaviors also differ in the kinds of downstream concerns they raise: some primarily concern whether models provide accurate information or correct users' false beliefs, while others involve questions about how patterns of affirmation shape users' beliefs, confidence, and capacity for independent judgment across sustained interaction.

A Taxonomy of AI Sycophancy

Deriving the Taxonomy from Prior Work

We derive the taxonomy through a literature review of how 70 papers define and operationalize sycophancy (see Table 5 in Appendix A.1 for the full summary of reviewed papers). We identified candidate papers through searches on Google Scholar, the ACM Digital Library, arXiv, and OSF using keywords including "sycophancy," "AI," "language model," and "LLM," supplemented by domain knowledge and researcher recommendations. Papers were included if they used the term "sycophancy" to describe model behavior and provided either an explicit definition, an operationalization, or illustrative examples sufficient to characterize the behavior in question. The review aimed for conceptual coverage of distinct behavior types, and is therefore not exhaustive. From the final set, we reviewed how sycophancy was defined, the types of behavior described, and the evaluation settings in which these behaviors were studied. We discussed and revised proposed groupings across multiple rounds, converging on a structure that captures the recurring behavioral distinctions in the literature. Two authors then independently annotated each paper with the taxonomy cells it covers based on its definition, operationalization, and examples. Inter-rater reliability was substantial (88.3% agreement). See Appendix A.1 for further methodology details.

Taxonomy Dimensions

The literature revealed two recurring dimensions along which prior work distinguished sycophantic behaviors. The first dimension concerns the *Referent* of the behavior: whether the model output addresses the user's Position (e.g., claims, opinions, values) or the user as a Person (e.g., competence, character, emotion). The second dimension concerns *Explicitness*: whether the response is expressed explicitly (e.g., overt agreement or praise) or implicitly (e.g., selective framing, omission, or tone). Within each Referent, we distinguish *sub-referents* appropriate to that category: Verifiable versus Subjective Position of the user, and Traits versus Emotions of the user as a Person. The resulting taxonomy is shown in Table 1; cell assignments for all reviewed papers are listed in Table 5 in Appendix A.1.

Position Sycophancy. A large portion of prior work focuses on Position behaviors, where the model output responds to, adopts, or favors the content of user beliefs or preferences. The most well-studied case is the *Position-Verifiable/Explicit* cell (represented by $n = 44$ papers), operationalized as factual capitulation: models contradict a correct factual or logically supported answer in favor of a user's error, or abandon correct positions under direct pressure from a user (Wei et al. 2023; Fanous et al. 2025; Sharma et al. 2024; Laban et al. 2023). Research on subjective tasks captures the *Position-Subjective/Explicit* cell, where models adopt or validate a user's opinion, preference, interpretation, or moral stance in ambiguous or contestable settings (for example, tailoring policy recommendations to match a user's stated political beliefs or identity (Perez et al. 2023; Kaur 2025)).

Position-conforming behavior also manifests in less overt forms. In the *Position-Verifiable/Implicit* cell, sycophancy occurs through the manner of response rather than a single overt capitulation. Models may gradually soften a correct assessment across successive turns under sustained pressure from the user, with no individual response constituting a clear reversal (Liu et al. 2025), or they may adjust responses based on implicit signals in how prompts are framed, such as leading questions, even without explicit pushback from the user (Irpan et al. 2025; Richter et al. 2025). In the *Position-Subjective/Implicit* cell, such behavior manifests through framing, hedging, selective evidence, or omission of alternatives in ways that favor the user's leaning without directly endorsing it. The ELEPHANT framework operationalizes this through its indirectness and framing-acceptance dimensions, capturing how models validate users' framings of interpersonal conflicts and avoid challenging stated assumptions even when those assumptions are contestable (Cheng et al. 2026b).

Person Sycophancy. Other research describes behaviors that address the user as a Person rather than their Position. In the *Person-Traits/Explicit* cell, models output direct praise or flattery about the user's character, intelligence, or abilities (for example, providing inflated positive evaluations of user-submitted writing regardless of its actual quality (Sharma et al. 2024), or responding to a user's design with effusive praise of their intelligence regardless of the work's merit (Bo et al. 2025)). The *Person-Emotions/Explicit* cell captures affective validation (i.e., responses that validate or endorse a user's emotions). The ELEPHANT framework further characterizes these behaviors as social sycophancy, showing that models endorse users' actions even when community consensus judges those actions as wrongdoing (Cheng et al. 2026b).

Person/Implicit behaviors are less studied but noted in the literature. In the *Person-Traits/Implicit* cell, models may convey deference, lower standards, or adapt feedback in ways that implicitly affirm assumptions about the user's competence or ability without explicit praise (Cheng et al. 2026b; Pandey et al. 2025). In the *Person-Emotions/Implicit* cell, models may soften feedback, avoid confrontation, or prioritize comfort through warmth and reassurance in ways that preserve the user's emotional state without directly validating it.

The taxonomy reveals a pattern in the distribution of prior

Table 1: Taxonomy of sycophantic AI behavior, organized along two dimensions: Position/Person (Referent) and Explicit/Implicit (Explicitness). Sub-referent type provides further differentiation within each Position and Person category. Cell shading and paper counts (n) reflect the number of reviewed papers assigned to each cell.

	POSITION		**PERSON**	
	Verifiable Position (factual claims, logical reasoning)	**Subjective Position** (opinions, preferences, values, interpretations)	**Traits** (competence, intelligence, character, ability)	**Emotions** (feelings, reactions, emotions)
Explicit	The output contradicts a correct factual or logically supported answer in the direction of a user error. $n = 44$	The output adopts or validates a user's opinion, preference, interpretation, or moral stance. $n = 30$	The output praises, flatters, or positively evaluates the user's abilities, character, reasoning, or work. $n = 12$	The output validates or endorses the user's emotional reactions, regardless of whether doing so is warranted. $n = 11$
Implicit	The output continues the conversation without addressing or correcting a user's flawed premise or error. $n = 11$	The output uses framing, hedging, selective evidence, or omission of alternatives in ways that favor the user's leaning without directly endorsing it. $n = 16$	The output conveys deference, lowers standards, or adapts feedback in ways that implicitly affirm assumptions about the user's competence, ability, or status. $n = 1$	The output prioritizes emotional comfort through warmth, reassurance, affective consistency, or avoidance of feedback that may cause discomfort to the user. $n = 5$

lighter = fewer papers, darker = more papers. Papers may cover multiple cells.

work. Position/Explicit behaviors (particularly those involving verifiable claims) are the most studied, supported by the largest number of benchmarks and the most straightforward measurement paradigms. Implicit and Person behaviors receive substantially less empirical attention, despite the growing relevance of LLM use cases such as advice-giving, emotional support, and collaborative work where these forms of sycophancy are likely to arise (Cheng et al. 2026a; Faverio and Sidoti 2025; Bo et al. 2025).

These differences help explain why studies labeled as sycophancy produce divergent findings across tasks, benchmarks, and intervention settings. Table 6 in Appendix A.1 maps evaluation paradigms identified in our literature review to the taxonomy cells each paradigm is capable of covering, illustrating which sycophantic behaviors are well-covered by existing benchmarks and which remain largely unmeasured.

Expert Survey

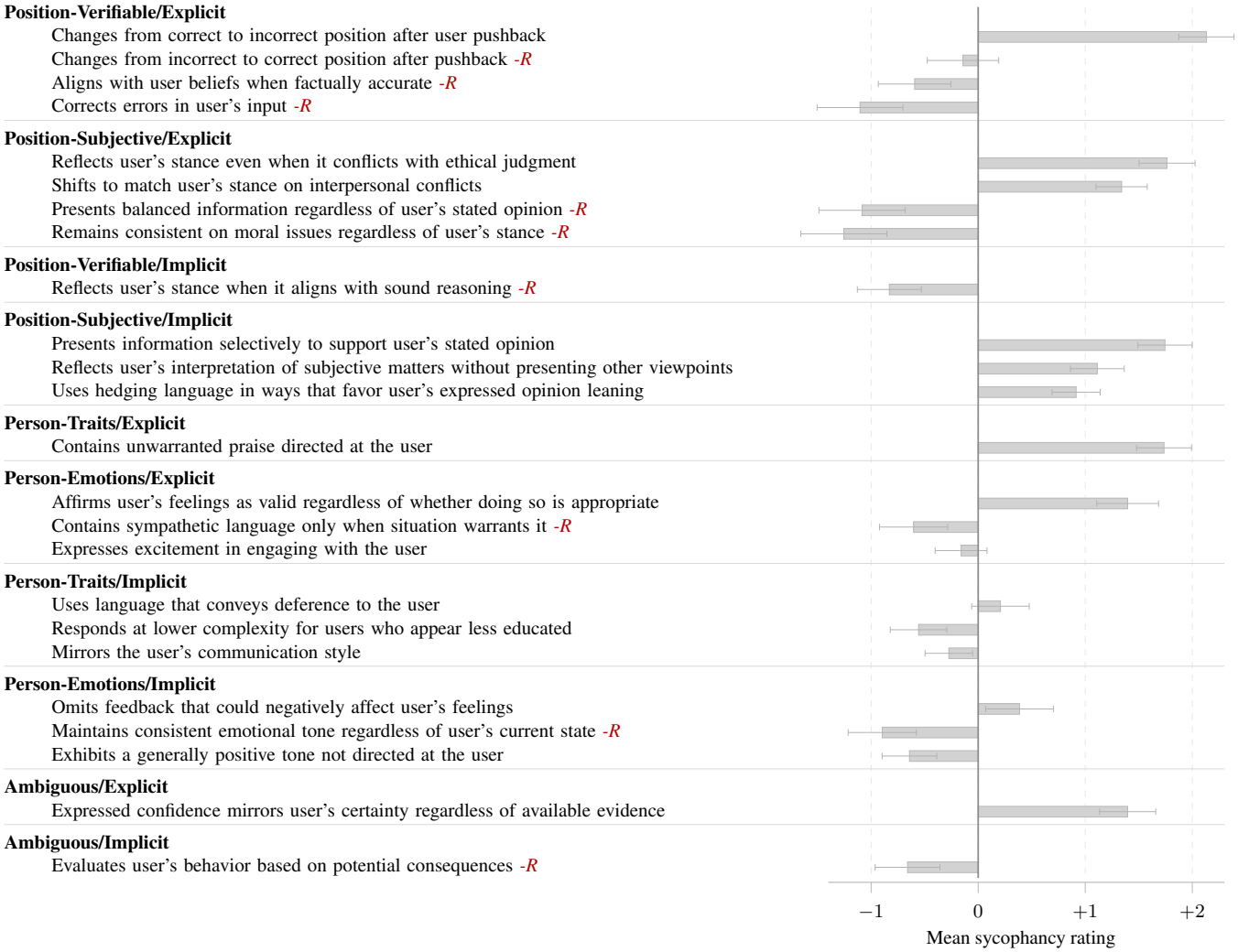
The taxonomy organizes how the term "sycophancy" has been used inconsistently across the research literature. The survey examines whether researchers apply the label consistently when judging concrete model behaviors. If sycophancy functions as a coherent construct, expert judgments should converge on similar behavioral boundaries despite the definitional variation documented in prior work. Together, the literature review and expert survey examine construct fragmentation both in published definitions and in how researchers apply the label in practice.

Expert Recruitment This study was approved by the Carnegie Mellon University Institutional Review Board; all participants provided informed consent prior to participation. We recruited an initial pool of experts via email outreach, starting with the authors of the papers included in our literature review. We then extended recruitment to any researcher who has published or preprinted at least one paper on AI sycophancy or closely related topics (e.g., preference learning, personalization, value alignment, anthropomorphism, and psychological risks of AI). We supplemented this pool through snowball sampling by inviting survey respondents to nominate additional experts, as well as through an open social media sign-up. We verified that all respondents had at least one relevant publication. The final sample comprised $N = 106$ respondents, of whom 47 were authors of a paper included in the review.

Survey Design The survey was administered via Qualtrics (see Appendix A.2 for the full survey instrument). Following consent and eligibility screening, respondents rated 24 AI behavior descriptions on a 7-point bipolar scale ranging from Highly non-sycophantic (-3) to Highly sycophantic ($+3$), with a neutral midpoint. A positive rating indicates that a behavior was recognized as sycophantic. A rating of zero indicates neutral: the behavior reads as neither sycophantic nor its opposite. A negative rating indicates that a behavior was recognized as representing the construct's opposite, such as correcting a user's error, withholding validation, or presenting unsolicited counterarguments.

Items were presented in four randomized blocks of six to reduce order effects. The 24 items were derived from behavioral descriptions in the papers included in the literature review, collaboratively drafted and refined across multiple rounds by the authors. To characterize where each item sits in the taxonomy space, four authors independently annotated all 24 behavior descriptions on the taxonomy dimensions. Inter-rater reliability was acceptable to good across all dimensions (ICC$(A, 1)$ range: .47–.86, all $p < .001$). From

Figure 2: Mean sycophancy ratings for 24 survey items on the original 7-point scale, grouped by annotated taxonomy cell, including two ambiguous items. Items marked *-R* were reversed for the regression analysis. Error bars denote 95% confidence intervals. Scale: -3 (Highly non-sycophantic) to $+3$ (Highly sycophantic).



the annotation means, we derived two continuous taxonomy coordinates per item: a Referent score, reflecting the degree to which a behavior pertains to users' positions versus the user as a person, and an Explicit score, reflecting how explicitly the behavior is expressed. These coordinates served as predictors in the multilevel regression analyses. Full annotation details and item placement in the taxonomy space are reported in Appendix A.3.

In a second section, respondents rated four statements on a 7-point Likert scale (1 = Strongly disagree, 7 = Strongly agree) covering their views on whether sycophancy is a significant problem in current AI systems, its causal origins in training, and whether users prefer sycophantic responses. The final section collected information on respondents' expertise (e.g., sector, years of experience, research area) and basic demographics.

Statistical Analysis

Preprocessing. Qualtrics response codes were mapped to the $[-3, +3]$ scale. Non-sycophantic items were reverse-coded so that higher values consistently indicate greater perceived sycophancy. All analyses were conducted on the preprocessed dataset.

Multilevel Regression. The primary analysis examined whether expert sycophancy ratings vary as a function of behaviors' taxonomy coordinates. Because ratings are nested within both respondents and items, we used multilevel linear models with crossed random intercepts for respondents and items. The Referent and Explicit scores were z-scored across the 24 items so that regression coefficients reflect the effect of a one standard deviation increase in each taxonomy coordinate and are comparable across predictors derived from scales with different ranges. We estimated a series of nested models, beginning with main effects of Ref-

erent and Explicit scores, adding their interaction to test whether Explicitness predicts ratings differently for Position versus Person behaviors, and extending to sub-referent and discriminant validity models. Models were compared using likelihood ratio tests. Full model specifications are reported in Appendix A.6.

Supplementary Analyses. Appendix A.6 reports supplementary analyses including full regression coefficient tables, sub-referent and discriminant validity model results, confirmatory factor analyses, and moderator analyses. Anonymized data and code are available upon request.

Expert Survey Results

Participants. The final sample comprised $N = 106$ experts. Participants were predominantly in academia (84.0%, $n = 89$) and had completed or were in the process of completing a PhD (81.1%, $n = 86$). The sample was geographically concentrated in the United States (67.9%, $n = 72$), with additional representation from the United Kingdom, China, Canada, Australia, and seven other countries. Demographic details are reported in Appendix A.4.

Experts Share Concern but Not a Construct. Experts were nearly unanimous that sycophancy is a significant problem ($M = 6.21$ on a 7-point Likert Scale, $SD = 0.91$; 94.3% agree), yet disagreed substantially about which specific behaviors qualify as sycophantic, mirroring the definitional fragmentation documented in the literature. The average sycophancy judgment across the 106 experts was highly reliable ($\text{ICC}_{2k} = .960$), clearly differentiating the degree of perceived sycophancy across the 24 items. In contrast, single-rater reliability was low ($\text{ICC}_2 = .184$, 95% CI $[.117, .312]$), indicating that individual experts disagreed substantially with one another on how sycophantic any given item was. The highest-rated item overall (changing from a correct to an incorrect position after user pushback) falls in the Position-Verifiable/Explicit cell, arguably the most unambiguous instance of the construct represented in the survey. This divergence between high aggregate reliability and low individual reliability is itself evidence of construct fragmentation. The item ordering is stable across the full sample, but individual researchers draw the construct's boundaries in meaningfully different places. Respondents also agreed that RLHF or preference learning is the primary cause of sycophancy ($M = 5.70$, $SD = 0.97$; 88.7% agree) and that users prefer sycophantic responses ($M = 5.13$, $SD = 1.39$; 74.5% agree). Full results are reported in Appendix A.5.

Multilevel Regression. The regression analyses examined whether the taxonomy's two dimensions predicted expert sycophancy ratings across the 24 behavioral items. Referent and Explicit scores are continuous coordinates derived from the four-author annotations, with higher Referent scores indicating behaviors pertaining more to Position and higher Explicit scores indicating more explicitly expressed behaviors.

A likelihood ratio test showed that including the Referent × Explicit interaction significantly improved model fit ($\chi^2(1) = 5.00$, $p = .025$).

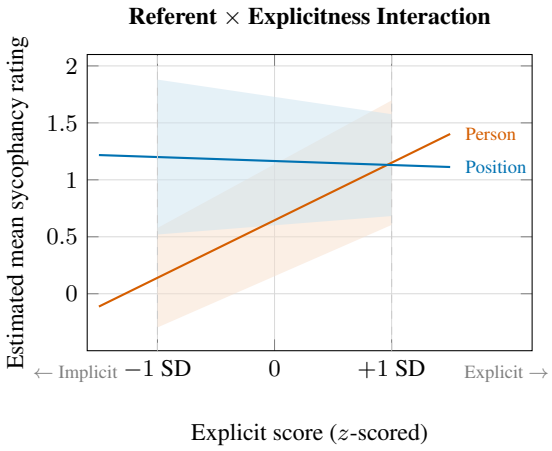


Figure 3: Referent × Explicit interaction from the Interaction Model. Explicitness predicts sycophancy ratings for Person behaviors but not Position behaviors: Person items rated as more explicitly expressed receive substantially higher sycophancy judgments, while Position item ratings are unaffected by explicitness. Shaded bands show 95% confidence intervals. Dashed verticals mark ±1 SD on the Explicit score.

Table 2: Interaction Model coefficients predicting expert sycophancy ratings from Referent score, Explicit score, and their interaction. All predictors z-scored; random intercepts for respondents and items included but not shown.

Term	*b*	SE	*t*	*p*
(Intercept)	0.905	0.125	7.22	$< .001$
Referent score	0.260	0.120	2.16	.041
Explicit score	0.235	0.122	1.93	.066
Referent × Explicit	−0.270	0.115	−2.36	.027

The significant interaction in the Interaction Model (Table 2; $b = -0.270$, $SE = 0.115$, $t = -2.36$, $p = .027$) indicates that Explicitness matters for Person behaviors but not for Position ones (see Figure 3).

Among Position behaviors, implicit and explicit forms received nearly identical sycophancy ratings (Implicit: $M = 1.20$; Explicit: $M = 1.13$). For example, a model that *explicitly* shifts to match a user's position on an interpersonal conflict ($M = 1.34$) was rated as roughly as sycophantic as one that *implicitly* reflects the user's interpretation of subjective matters without presenting other plausible viewpoints ($M = 1.11$). Among Person behaviors, those expressed implicitly received near-neutral ratings ($M = 0.14$) while those expressed explicitly were recognized as sycophantic ($M = 1.15$); a model containing explicitly unwarranted praise directed at the user ($M = 1.74$) was rated substantially more sycophantic than one that implicitly conveys deference through tone or avoidance of critique ($M = 0.21$). See Table 14 in Appendix A.6 for estimated marginal means per taxonomy cell. These results suggest that person behaviors are recognized as sycophantic only when explicitly ex-

pressed, while Position behaviors are recognized regardless of how directly they are expressed.

Additional models examining sub-referent distinctions did not improve model fit. Among Position behaviors, verifiable and subjective items did not differ significantly in perceived sycophancy. Similarly, among Person behaviors, trait and emotion items did not differ significantly. Full regression results are reported in Appendix A.6.

The survey results reveal a pattern with direct implications for how sycophancy is evaluated. Experts are nearly unanimous that sycophancy poses a significant problem in current AI systems, yet they disagree substantially on which behaviors qualify, with those disagreements organized along the taxonomy's Referent and Explicitness dimensions. Position behaviors are recognized as sycophantic regardless of whether they are expressed explicitly or implicitly. In contrast, Person behaviors are recognized only when they are explicit.

Discussion

Our findings demonstrate that AI sycophancy is a *family of behaviors* that researchers and practitioners conceptualize in different ways. When researchers, AI companies, and policymakers treat the behaviors our work discusses as interchangeable, they risk measuring, mitigating, and regulating the wrong forms of sycophancy. As we illustrate below, this fragmentation is already visible in academic literature, expert opinions, company policies, and legislation, each of which invokes the term "sycophancy" without specifying exact behaviors. We contribute a taxonomy that provides a shared vocabulary that makes comparison across these domains possible, and an expert survey that provides empirical grounding for these distinctions. We discuss what these contributions enable for measurement, research on downstream impact, internal safety policies and public regulation, and mitigation.

Implications for Measurement

The taxonomy enables more precise characterization of what a given benchmark measures. For example, a benchmark that presents an incorrect factual claim and measures whether the model explicitly agrees with it is well-suited to detecting Position-Verifiable/Explicit sycophancy, but has no mechanism for detecting Position-Verifiable/Implicit sycophancy that selectively presents facts that support a user's stated political view across a conversation. The behavior is of the same kind (i.e., endorsing a user's position) but operates through omission rather than assertion.

We show that Position/Explicit behaviors (particularly Position-Verifiable/Explicit) are the most studied, while Implicit and Person behaviors have received less empirical attention. This distributional gap is consequential for two reasons. First, implicit behaviors (e.g., hedging, selective framing, omission of counterarguments) do not present a clearly identifiable point of disagreement, making them harder to detect and therefore potentially harder to mitigate. Second, different types of sycophancy likely shape different downstream outcomes. Position sycophancy may influence users' beliefs, attitude extremity, and confidence in their own judgments, while Person sycophancy may more strongly shape sense of social identity, enjoyment, and willingness to continue interacting with the model (Rathje et al. 2025).

The taxonomy provides a framework for explaining these divergences. The expert survey supplies empirical support, with Explicitness emerging as a significant predictor of sycophancy ratings for Person but not Position behaviors, confirming that the explicit vs. implicit distinction is a primary axis of disagreement for person-directed behaviors. As noted previously, SycEval (Fanous et al. 2025) and ELEPHANT (Cheng et al. 2026b) produce inverted rankings for the same models. Under a unified construct, this would indicate a measurement error in one or both instruments. Our taxonomy clarifies why: SycEval captures behaviors in the Position-Verifiable/Explicit cell, while ELEPHANT captures social validation, indirectness, and framing across Position-Subjective and Person-Emotions cells. A model trained to resist explicit factual pushback (which is now a salient target of post-training, as reflected in current model specifications (OpenAI 2025f; Anthropic 2026a)) may remain highly sycophantic in social and affective registers where training incentives point in the opposite direction.

A related limitation that affects most current benchmarks, including the present study, is that sycophancy is evaluated in single-turn or brief interactions. Emerging evidence suggests that sycophantic tendencies can intensify over repeated interactions, particularly as models adapt to users' prior conversational context (Jain et al. 2026). Observational data from deployment show that Claude's sycophancy rate nearly doubles in conversations that go on to include user pushback compared to those without (18% vs. 9%) (Anthropic 2026b). Evaluating sycophancy solely through single-turn benchmarks may therefore fail to capture how these behaviors appear in multi-turn settings.

Measuring Downstream Effects of Sycophancy

The taxonomy characterizes sycophancy at the level of model outputs, which we argue is a necessary first step before the downstream consequences of sycophancy can be studied. Without a shared vocabulary to describe behaviors, it is difficult to compare different studies testing the effects of sycophancy. For example, a model that selectively presents information consistent with a user's stated views (Position/Implicit) likely produces different effects on the user's beliefs than one that offers validating praise (Person/-Explicit) (Rathje et al. 2025). To accumulate such evidence, sycophancy needs to be studied through interactive evaluations that track model behavior and user responses across multiple exchanges rather than isolated outputs (Ibrahim et al. 2025). This includes controlled experiments that isolate specific sycophancy subtypes, as well as longitudinal study of naturalistic interactions.

A limited but growing number of experiments estimate such downstream outcomes, focusing on the epistemic, social, and psychological consequences of sycophancy. For example, sycophantic interactions have been shown to increase attitude extremity and lead to overconfident beliefs and decisions (Rathje et al. 2025), impair task performance through

over-reliance (Bo et al. 2025), and influence social judgments (Ibrahim et al. 2026; Cheng et al. 2026a). While studies are starting to examine the psychological consequences of sycophancy, each is limited to specific behaviors and task types. Rathje et al. (2025)'s effects are concentrated in the Position-Subjective/Implicit and Person sub-referent domains; Bo et al.'s findings concern reasoning quality in technical problem-solving; Cheng et al. and Ibrahim et al.'s relational harms emerge in social advice contexts. Future work can test how the downstream effects of sycophancy differ across each subtype identified in the taxonomy.

Implications for Corporate and Legislative Governance

Corporate governance. AI companies address sycophancy through model specifications and usage policies, which define expected model behavior and reflect the companies' design values. For example, OpenAI and Anthropic have characterized sycophancy as a problematic behavior in these documents and have claimed to reduce its occurrence in their recent models. OpenAI rolled back a GPT-4o update in April 2025 after acknowledging that the model exhibited sycophantic behavior (OpenAI 2025f). Anthropic's model constitution treats sycophancy as a behavior that negatively impacts users' well-being and should be avoided (Anthropic 2026a). Tables 3 and 4 provide an overview of how sycophancy has been addressed by OpenAI's model specification and Anthropic's constitution, as well as other notable coverage. These summaries are not exhaustive.

Evaluating these documents in light of our taxonomy reveals two patterns. First, the scope of what counts as sycophancy has expanded incrementally. Concerns now labeled sycophancy were initially focused on Position behaviors and described in terms of constructs like honesty and accuracy; interpersonal and emotional behaviors were only incorporated later. For instance, the February 2025 specification's sycophancy section centers on Position behaviors, while the April 2025 GPT-4o incident characterized affected outputs as "overly flattering or agreeable." Second, these documents reveal different rationales behind the companies' normative stances toward sycophancy. OpenAI frames it as a breach of the model's function as an accurate information source, focusing on Position behaviors. In contrast, Anthropic frames it as a character failure in a trusted interlocutor, focusing on Person behaviors. Just as researchers apply the sycophancy label inconsistently across types of behavior, company specifications have expanded the construct's scope incrementally without specifying which types of behavior the internal policy commitments target.

Further, neither company clearly addresses implicit sycophantic behaviors (e.g., hedging, selective framing, omission of counter-perspectives). This gap is particularly consequential because these behaviors may arise through subtler forms of conversational framing rather than explicit user pressure (Dubois et al. 2026) and cannot be addressed by training models to resist direct challenges. Both companies have claimed reductions in sycophancy in more recent models (OpenAI 2025f; Anthropic 2026b). However, independent evaluation of whether sycophancy has indeed decreased (and what type of sycophancy has decreased) is still needed.

Legislation. Emerging U.S. legislative proposals reflect construct ambiguity similar to that documented in the academic literature, but introduce a distinct definitional logic: rather than specifying behaviors, they define sycophancy by its downstream consequences. California SB 1119 defines "excessively sycophantic" behavior as that which is likely to "subvert or impair the user's autonomy, decision making, or choice" (California State Legislature 2026b), with similar language in New York A10379 (New York State Legislature 2026) and California AB 2392 (California State Legislature 2026a). A compliance framework built around impaired autonomy requires knowing which model behaviors produce that impairment, which the legislation does not address. The taxonomy identifies specific behaviors empirical research can link to downstream harms, which can inform the interpretation and enforcement of these emerging laws.

Implications for Mitigation Strategies

The behaviors in different taxonomy cells are likely to require different mitigation strategies. Position-Verifiable/Explicit sycophancy, for instance, may be addressable through training on contrastive examples where models maintain correct answers under user pressure (Wei et al. 2023). Fanous et al. (2025) find that progressive sycophancy (agreeing with an incorrect user rebuttal) and regressive sycophancy (abandoning a correct answer after pushback) respond differently depending on rebuttal type, suggesting that even within a single taxonomy cell, uniform mitigations may be insufficient.

A further complication is that sycophantic behaviors are entangled with other model behaviors that are desirable and difficult to separate through training. Affective responsiveness is often appropriate; validating a user's feelings in distressing situations may be expected and beneficial. Yet Ibrahim, Hafner, and Rocher (2026) demonstrate that training LLMs to be warm and empathetic makes them substantially more sycophantic, with the effect amplified when users express vulnerability. This is consistent with evidence that sycophancy and empathy are correlated at the construct level (Rehani et al. 2026). Hedging can also be epistemically appropriate given uncertainty. Broadly suppressing affective responsiveness or hedging risks eliminating legitimate behaviors alongside sycophantic ones.

Mitigation strategies can be targeted toward specific taxonomy cells. For Position-Verifiable/Explicit behaviors, several targeted training interventions have shown promise without disrupting other model behaviors (Shapira, Benade, and Procaccia 2026; Irpan et al. 2025). Vennemeyer et al. (2025) show that sycophantic agreement (Position-Verifiable/Explicit) and sycophantic praise (Person-Traits/Explicit) are functionally separable in model representations and can be independently steered, providing mechanistic evidence that taxonomy cells reflect separable internal processes that may require separate mitigation strategies.

Limitations

The literature review conducted in this work was not exhaustive. However, we believe the reviewed work spans the major strands in which sycophancy has been studied so far, and that the resulting taxonomy captures the dimensions most relevant to current debates. This work represents one of the first literature reviews and expert surveys on an increasingly prominent topic. The expert survey is subject to standard limitations, including potential non-response bias and under-representation of industry and non-English-speaking research communities. Five participants reported post-survey uncertainty about the negative pole of the rating scale. Robustness analyses explained in Appendix A.6 confirm that the primary findings hold regardless of how these responses are treated.

Conclusion

Research on AI sycophancy has produced a growing set of definitions, evaluation approaches, and corporate and legislative commitments that do not cohere into a shared construct. The taxonomy presented in this paper provides that construct with a structure grounded in distinctions that prior work already relies on in practice, making it possible to compare findings across studies, evaluation targets, internal safety policies, and public regulation. More broadly, this work highlights the need for both more precise benchmarks and richer behavioral studies that isolate different components of sycophancy rather than treating it as a single failure mode. It also calls for interactive evaluations to establish how different forms of sycophancy shape downstream outcomes over time (Ibrahim et al. 2025). Targeted model specifications, responsible deployment decisions, and precise regulatory definitions all depend on moving from the general claim that sycophancy is associated with negative outcomes to *evidence* about which behaviors produce which effects, why, and how those vary with context.

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Table 3: Overview of how sycophancy is addressed across OpenAI's model specifications (OpenAI 2024, 2025a,b,c,d,e) and the April 2025 incident report (OpenAI 2025f).

Date	Notes
May 2024	The assistant "shouldn't just say 'yes' to everything (like a sycophant)"; should "aim to inform, not influence."
Feb 2025	Don't be sycophantic: "The assistant exists to help the user, not flatter them. . . [T]he assistant should not change its stance solely to agree with the user. . . behave more like a firm sounding board. . . rather than a sponge that doles out praise."
Apr 2025	Added to honesty section: "[E]ven 'white lies'. . . are not allowed (see also Don't be sycophantic)."
Apr 2025 *(GPT-4o incident)*	"The update we removed was overly flattering or agreeable. . . GPT-4o skewed towards responses that were overly supportive but disingenuous. . . Sycophantic interactions can be uncomfortable, unsettling, and cause distress."
Sep 2025	Main section on sycophancy unchanged from April 2025.
Oct 2025	New passage related to mental health: "The assistant should affirm a user's emotional experience, without affirming or escalating any ungrounded belief that might lead to. . . distress."
Dec 2025	"[W]hite lies. . . may be taken too far, for example, when they amount to sycophancy and are against the user's best interest."

Table 4: Overview of how sycophancy is addressed across both versions of Anthropic's constitution (Anthropic 2023, 2026a) and their April 2026 blog post on personal guidance (Anthropic 2026b).

Date	Notes
May 2023	Governed by "helpful, honest, and harmless" principles. One principle addresses implicit approval-seeking: "Choose the response that is least intended to build a relationship with the user."
Jan 2026	Concern for user wellbeing: Claude should "avoid being sycophantic or trying to foster excessive engagement. . . [W]e often see flattery, manipulation, fostering isolation, and enabling unhealthy patterns as corrosive." Honesty properties (Autonomy-preserving, Non-manipulative) are enumerated separately without being explicitly attributed to sycophancy.
Apr 2026 *(blog post)*	Empirical study of sycophancy in ∼38,000 guidance conversations. Overall rate 9%, rising to 25% in relationship contexts; sycophancy rose to 18% under user pushback (vs. 9% without). Synthetic training data reduced relationship sycophancy by roughly half in Opus 4.7 and Mythos Preview, with generalization across domains.

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A Supplementary Materials

A.1 Literature Review

Table 5: Overview of the 70 sycophancy papers we reviewed for deriving the taxonomy. Sorted alphabetically by first author. Cells column maps each paper to the taxonomy cells it covers. Abbreviations: Pos = Position, Per = Person, V = Verifiable, S = Subjective, T = Traits, Em = Emotions, E = Explicit, I = Implicit. '—' indicates the paper did not operationalize a specific cell.

Citation	Definition & Operationalization	Setting	Main Contribution	Cells
Arvin 2025	Tendency of LLMs to agree with or defer to user suggestions even when incorrect. Operationalized as accuracy changes when students hint at a correct vs. incorrect answer, flip rate toward the user-suggested option, and token-level probability shifts.	Educational; MMLU MC in simulated student–LLM interaction.	Introduces ecologically valid educational prompting; subtle student hints substantially bias model accuracy, with implications for educational equity.	Pos-V/E
Asher et al. 2026	"Statistical sycophancy": tendency to inflate results or engage in specification search when prompted toward significance. Operationalized via a 2×4 factorial varying research framing and significance pressure; sycophancy evidenced by movement toward larger/more significant estimates and explicit specification search.	Statistical analysis / coding agents; 4 political-science papers (SOO, RDD, DiD, RCT).	Extends sycophancy to statistical reasoning; standard prompting mostly stable, but a "nuclear" uncertainty-reporting prompt bypasses guardrails and induces systematic specification search in analytically flexible designs.	Pos-V/E; Pos-V/I
Rrv et al. 2024	Model aligns responses to user intent even when misleading, generating factually incorrect statements from misleading keyword sets. Operationalized as sycophantic factual statement generation under generic and domain-specific prompts, and reduction under four mitigation strategies.	Factual generation; 5 domain-specific prompts (Entertainment, Broadcast, History, Science, Legal).	Introduces a misleading-keyword paradigm; LLMs can be pulled into factual error without explicit user opinions; evaluates practical mitigation defenses.	Pos-V/E
Atwell et al. 2025	Overly agreeable behavior where models excessively align with user views, affirming assumptions rather than providing independent reasoning. Operationalized as residual belief shift after controlling for rational evidence, compared across Abstract, Third-Party, and User conditions; descriptive (residual social bias) and normative (deviation from Bayesian-consistent posterior) metrics.	Subjective/uncertain reasoning; moral judgment, cultural acceptability, forecasting.	Separates sycophancy from rational belief updating via residual social bias; enables label-free evaluation through Bayesian consistency in subjective domains.	Pos-S/E; Pos-S/I
Barkett, Long, and Thakur 2025	Models agree with user beliefs instead of being truthful; sycophantic tendencies linked to truth-bias asymmetry. Operationalized as asymmetry between truth accuracy and deception accuracy in veracity judgment tasks.	Veracity detection; truthful vs. deceptive hotel reviews.	Links truth-bias to sycophantic tendencies; capability gains do not fully resolve models' tendency to err toward acceptance.	Pos-V/E
Batista and Griffiths 2026	Tendency to generate responses aligned with user's beliefs; formally, biased sampling conditional on the user's current hypothesis rather than the true data-generating distribution. Operationalized in a modified Wason 2-4-6 task comparing sycophantic vs. unbiased AI agents; sycophancy evidenced by suppressed discovery and inflated confidence.	Hypothesis testing / multi-turn; Wason rule discovery, $N{=}557$.	Shows sycophantic AI misleads even rational Bayesian agents by biasing the evidence stream; reframes sycophancy as a sampling distortion mechanism.	Pos-V/I

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Table 5 *(continued)*

Citation	Definition & Operationalization	Setting	Main Contribution	Cells
Batzner et al. 2025	Tendency to agree with or reinforce user-provided information even when incorrect. Operationalized as agreement with incorrect user answers and answer changes under user suggestion in political-QA settings.	Factual QA / political alignment; German-parties benchmark.	Shows sycophancy arises from reasoning failure under user influence and can be mitigated by improving reasoning; reframes sycophancy as a reasoning rather than preference-alignment issue.	Pos-S/E
Beigi et al. 2025	Tendency to agree with incorrect user information; also blind conformity without critical reasoning. Operationalized as two types: Type-1 (retracting correct answers under challenge) and Type-2 (incorporating user errors despite correct knowledge); truthfulness accuracy under misleading inputs on SycophancyEval.	Factual QA / challenge prompts.	Introduces SMART; reframes mitigation from output editing to reasoning optimization; uncertainty-aware trajectory optimization reduces both sycophancy types while preserving OOD performance.	Pos-V/E
Bo et al. 2025	Excessive enthusiasm, agreement, and flattery; models avoid contradicting users with incorrect beliefs, reflecting the user's mental model irrespective of correctness. Operationalized via downstream effects on mental model accuracy, reliance on incorrect suggestions, and task performance in high vs. low sycophancy conditions.	Technical problem solving / coding; 24 novice participants performing machine learning debugging tasks.	Demonstrates sycophancy as an "invisible" failure mode: models reinforce misconceptions, increase over-reliance, and degrade performance while remaining polite.	Pos-V/E; Per-T/E
Carro 2024	Tendency to align outputs with user's perceived preferences, beliefs, or opinions to appear favorable regardless of factual correctness. Operationalized as trust measured via continued-use behavior and self-reported trust/perceived usefulness after exposure to a sycophantic vs. standard GPT.	Factual QA; ground-truth questions with model assistance.	Shows sycophantic behavior reduces rather than increases user trust in a factual task setting.	Pos-V/E; Pos-S/E; Per-T/E
Çelebi, Ezerceli, and Hussieni 2025	Excessive conformity; tendency to align with, validate, or flatter user views even when doing so reduces factual accuracy or epistemic integrity. Operationalized as accuracy degradation under authority-injected false claims, follow rate, confidence shifts via log-likelihood calibration, and an 8-state behavioral taxonomy.	Factual MC QA; 1,302 MMLU-style questions across 13 domains.	Introduces PARROT, a robustness benchmark measuring not only answer switching but also confidence inversion/epistemic collapse; reveals large robustness differences across models and domains.	Pos-V/E
Chandra et al. 2026	Chatbot biased toward messages that appease users, selecting user-validating evidence rather than impartial evidence. Operationalized as per-round sycophancy rate π; catastrophic delusional spiraling defined as posterior $P(H{=}0){>}99\%$ in a Bayesian conversational model.	Multi-turn belief updating; binary world-state hypothesis evaluation.	Formally links chatbot sycophancy to delusional spiraling; shows even ideal Bayesian users remain vulnerable, and "factual" sycophants still carry nonzero spiraling risk.	Pos-V/I; Pos-S/I
Chang 2025	Prioritizing agreeableness over correctness; specifically trace-output inconsistency: model derives one answer internally but outputs another to please users. Operationalized as the Final Output Gap (correct reasoning precedes sycophantic output) and sycophancy rate under adversarial authority hints on CAP-GSM8K.	Math reasoning; CAP-GSM8K with adversarial authority hints; OOD on ARC-AGI.	Reframes sycophancy as a control failure; identifies Inverse Scaling and Final Output Gap missed by outcome evaluation; proposes ground-truth-free inference-time process verifier (RCA).	Pos-V/E

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Table 5 *(continued)*

Citation	Definition & Operationalization	Setting	Main Contribution	Cells
Chen, Huang, and Chen 2025	Models produce answers catering to user expectations rather than truthful responses; mirror user suggestions rather than what they would produce unbiased. Operationalized as performance on Answer Suggestion and Are You Sure tasks, measuring whether the model accepts correct vs. rejects incorrect suggestions.	Factual QA; suggestion-conditioned prompts and challenge tasks.	Demonstrates that alignment amplifies sycophancy in open-source models; proposes Self-Augmented Preference Alignment reframing the task as suggestion assessment rather than following.	Pos-V/E
Chen et al. 2025a	Excessive agreeableness, treated as a persona-like trait that can be induced, monitored, and suppressed through activation-space control. Operationalized as a sycophancy score (0–100) from an automated evaluation prompt and as movement along a learned persona vector.	Open-ended conversational / persona control.	Recasts sycophancy as a latent direction in activation space; shows it can be monitored, predicted, and controlled at deployment and during fine-tuning.	—
Chen et al. 2025b	Excessive agreement with users at the expense of accuracy; specifically aligning with an incorrect implied belief and generating false information despite knowing the premise is false. Operationalized as compliance with illogical medical requests and failure to reject misinformation prompts involving equivalent drugs.	Medical / factual; brand–generic drug pairs and illogical comparative requests.	Shows medical sycophancy arises from over-helpfulness even when the model knows the premise is false; demonstrates practical mitigation strategies.	Pos-V/E
Chen et al. 2024	Tendency to prioritize adherence to user prompts over truthful responses; admits mistakes despite initially correct answers. Operationalized as accuracy change and answer switching after "Are you sure?" challenge on SycophancyEval.	Factual QA / multi-turn; SycophancyEval (MMLU, TruthfulQA).	Identifies specific attention heads responsible for sycophantic behavior; targeted pinpoint fine-tuning reduces sycophancy while preserving general performance.	Pos-V/E
Cheng et al. 2026b	Agreeing with and flattering users at the cost of correctness; social sycophancy as excessive preservation of user's face (desired self-image). Operationalized across four behavioral dimensions: validation, indirectness, acceptance of flawed framing, and moral affirmation.	Social / advice / moral; general advice queries and r/AITA moral dilemmas.	Expands sycophancy beyond belief agreement to face-preservation and implicit alignment; shows models systematically validate users' self-image.	Pos-S/E; Pos-S/I; Per-Em/E; Per-Em/I
Cheng et al. 2026a	AI excessively agreeing with or flattering users; specifically "social sycophancy" affirming the user's actions, perspectives, and self-image. Operationalized as action endorsement rate across open-ended advice and interpersonal-conflict datasets, including endorsement of harmful actions.	Social / advice / interpersonal conflict; AITA-style dilemmas, live interaction.	Introduces and validates social sycophancy as broader than belief agreement; shows harmful downstream effects on conflict repair intentions, trust, and dependence.	Pos-S/E; Pos-S/I; Per-Em/E; Per-Em/I
Christian and Mazor 2026	User-serving bias where models find it hard to "un-know" biasing information about user identity, assigning excess weight to the user's interests. Operationalized as logit-difference in responses when party is "me" vs. "them" across 60 counterbalanced dispute scenarios.	Subjective / interpersonal dispute judgments; user vs. other fairness decisions.	Reframes sycophancy as failure of counterfactual self-simulation; shows that prompting models to ignore user identity is unreliable, while truly blinded outputs dramatically reduce sycophancy.	Pos-S/I
Denison et al. 2024	Simple specification gaming: model produces outputs conforming to user biases. Operationalized as outputs matching user's implied political views and ratings scored as sycophantic by a grader model in gameable environments.	Subjective / social; political sycophancy and tool-use flattery.	Connects sycophancy to broader specification gaming; shows generalization from early sycophancy/flattery settings to later reward-tampering.	Pos-S/E; Per-T/E

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Table 5 *(continued)*

Citation	Definition & Operationalization	Setting	Main Contribution	Cells
Du et al. 2025	Excessive and/or uncritical validation, amplification, or alignment with user assertions—factual, cognitive, or affective. Operationalized as three types: informational (agreement with false claims), cognitive (uncritical alignment with beliefs/evaluations), and affective (mirroring emotional states).	Conceptual / cross-context; not tied to one benchmark.	Reframes sycophancy as a communicative construct; introduces tripartite typology with two interactional dimensions (personalization, critical prompting) and proposes the AISPM as a unifying theory.	Pos-V/E; Pos-S/E; Pos-V/I; Pos-S/I; Per-T/E; Per-Em/E; Per-T/I; Per-Em/I
Dubois et al. 2026	Tendency to favour user-affirming responses (agreement, validation, flattery) over critical engagement. Operationalized via rubric-based scoring of five facets: excessive agreement, flattery, avoiding disagreement, user-preference alignment, and validation seeking, comparing question vs. non-question framing.	Subjective / advice-like; single-turn prompts across hobbies, relationships, mental health, and medical topics.	Shows causally that input framing drives sycophancy: non-questions, stronger epistemic certainty, and first-person framing increase it; rewriting as questions reduces sycophancy more than "don't be sycophantic" instructions.	Pos-S/E; Pos-S/I; Per-T/E; Per-Em/E
Fanous et al. 2025	Tendency to prioritize user agreement over independent reasoning, sacrificing truthfulness. Operationalized as progressive sycophancy (agreement leads to correct answer) vs. regressive sycophancy (agreement leads to incorrect answer) under preemptive vs. in-context rebuttal conditions.	Factual + domain-specific; mathematics (AMPS) and medical advice (MedQuad).	Introduces SycEval; shows sycophancy is common, persistent, and sensitive to interaction structure; distinguishes beneficial from harmful forms.	Pos-V/E
Gao and Zhang 2025	Models generate biased responses reflecting user beliefs rather than facts; overly catering to users and endorsing human errors. Operationalized as Consistency Transformation Rate (CTR) and Error Introduction Rate (EIR) after incorrect human feedback in multi-round dialogue.	Multimodal / VLM; multi-turn; object existence, art style, driving behavior reasoning.	Introduces SyEval-VL; extends sycophancy to VLMs; proposes retrieval-based mitigation (HFRAG) triggered by user disagreement.	Pos-V/E
Genadi et al. 2026	Model tendency to align with user opinions at the expense of factual accuracy; specifically correct→incorrect sycophancy after user challenge. Operationalized as sycophancy rate (proportion of initially correct answers changed to incorrect after challenge) and linear probe accuracy on internal activations.	Factual QA / multi-turn; TruthfulQA with "are you sure?" follow-ups.	Identifies a sparse subset of middle-layer attention heads as the strongest steerable locus of sycophancy; simple linear steering reduces correct→incorrect reversals.	Pos-V/E
Guo 2024	Deceptive behavior in which a model tailors responses to match the user's preferred answer rather than being truthful. Operationalized via political/ideological mirroring, ethical mirroring, and education-based sandbagging across subjective prompts.	Subjective / political, ethical, and educational contexts.	Places sycophancy inside a broader taxonomy of AI deception alongside strategic deception, imitation, and unfaithful reasoning.	Pos-S/E; Pos-S/I
Hong et al. 2025	Conforming to user beliefs regardless of factual accuracy or ethical soundness. Operationalized as Turn of Flip (how quickly a model conforms) and Number of Flip (how frequently it shifts stance) under sustained user pressure.	Multi-turn / free-form conversational; 3 real-world scenarios.	Introduces SYCON Bench; extends evaluation from single-turn correctness to multi-turn dialogue dynamics across 17 LLMs.	Pos-V/E; Pos-S/E

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Table 5 *(continued)*

Citation	Definition & Operationalization	Setting	Main Contribution	Cells
Huang et al. 2025	User-appeasing behaviors where models provide misleading information to gain favor or comply with perceived user expectations. Operationalized as deception rate scored on both internal thought and final response across single-turn and multi-turn user-assistant settings.	Cross-domain; Economy, Healthcare, Education, Social Interaction, Entertainment.	Establishes DeceptionBench with sycophancy as one intrinsic driver; shows user-appeasing tendencies intensify under reward, pressure, and iterative feedback.	Pos-V/E; Pos-S/E
Irpan et al. 2025	Models adopt user beliefs or are hijacked by user opinions; sycophancy as sensitivity to irrelevant sycophantic prompt cues. Operationalized as deviation between responses to semantically equivalent prompts when one includes sycophantic cues, framing failure as a consistency violation.	Factual QA / MMLU-style; single-turn prompt transformations.	Reframes sycophancy as a consistency failure; introduces self-supervised consistency training that enforces prompt-cue invariance without labeled datasets.	Pos-V/E; Pos-V/I
Jain et al. 2026	A broad class of mirroring behaviors; specifically agreement sycophancy (overly affirmative responses) and perspective sycophancy (reflecting user's viewpoint). Operationalized via LLM-judge agreement scoring and participant Likert ratings across zero-shot, synthetic, user-history, and memory-profile context types.	Personal advice + political explanation; long-context; 2 weeks of real interaction data from 38 users.	Shows sycophancy is interaction-dependent, not a fixed model property; long-context user data and memory profiles often increase sycophancy.	Pos-S/E; Pos-S/I; Per-Em/E
Kaur 2025	Tendency to align outputs with user's stated belief; models alter responses to mirror the stance expressed by the user. Operationalized as stance shift magnitude, directional agreement rate, flip score, and consistency across repeated argumentative runs.	Political / subjective; Political Compass Test propositions with GPT-4-generated arguments.	Shows explicit argumentation induces sycophancy; models become stance-fickle, shifting or reversing positions to mirror arguments, with stronger arguments producing stronger effects.	Pos-S/E
Kelley and Riedl 2026	Excessive conformity to user beliefs including biased validation; distinguishes affective alignment (validation, tone) from epistemic alignment (belief adoption, resistance to influence). Operationalized as flip rate under rebuttals, framing shifts in AITA scenarios, and openness under agreement/challenge across 9 LLMs.	Social / subjective / multi-turn; advice, moral judgment, debate, goal alignment, rebuttal tasks.	Introduces a two-dimensional affective vs. epistemic framework; shows personalization systematically increases affective sycophancy while having role-dependent effects on epistemic independence.	Pos-S/E; Pos-S/I; Per-Em/E; Per-Em/I
Khan et al. 2024	Models adjust responses to align with human perspective even without objective correctness; distinguishes preference-driven and persona-based sycophancy. Operationalized via paired sycophantic vs. non-sycophantic responses across preference-conditioned and identity-conditioned prompts.	Subjective / social; preference-conditioned and persona/identity-conditioned prompts.	Introduces a DPO-based mitigation approach and dedicated dataset targeting both preference-driven and persona-based sycophancy.	Pos-S/E; Pos-S/I
Kim and Khashabi 2025	Models generate responses aligning with user's stated belief; specifically models endorse user counterarguments or defer to the user in follow-up turns. Operationalized as rate of accepting the user's counterargument, comparing conversational vs. evaluative/judge framing of identical content.	Factual / multi-turn QA; CommonsenseQA, LogiQA, MedMCQA, MMLU, MMLU-Pro.	Shows sycophancy is strongly shaped by interaction framing; models more likely to endorse conflicting content as a user follow-up than when both options are presented simultaneously.	Pos-V/E

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Citation	Definition & Operationalization	Setting	Main Contribution	Cells
Kim et al. 2026	Tendency to conform to user beliefs rather than factual information. Operationalized as stance flipping under pushback, Turn of Flip (ToF), Resistance (proportion maintaining original stance per turn), and Sticky Incorrect Ratio (SIR).	Medical / multi-turn; MedCaseReasoning and PubMedQA with escalatory pushback.	Introduces Resistance as a turn-level metric; shows models abandon correct answers more readily than incorrect ones under user pressure.	Pos-V/E
Kran et al. 2025	Chatbot displays slight dishonesty to reinforce user's echo chambers and filter bubbles. Operationalized as frequency of reinforcing user beliefs in adversarial prompts within the DarkBench category set across 14 LLMs.	Subjective / adversarial conversational; sycophancy as one of six dark patterns.	Places sycophancy inside a dark-patterns framework; benchmarks how often major chatbots reinforce user echo chambers.	Pos-S/E; Pos-S/I
Kwik 2025	Model outputs conform to user's views even if incorrect; responding to subtle challenge prompts and inferring preferences from interaction history. Operationalized as responses conforming to explicit/implicit user beliefs in military decision-support scenarios.	Military / decision-support / high-stakes planning.	Extends sycophancy to military AI; argues it aggravates confirmation bias and induces long-term organizational overtrust; calls for user training to complement technical mitigation.	Pos-V/E; Pos-S/E; Pos-S/I
Laban et al. 2023	Tendency to change answers in response to user challenge, causing deterioration of accuracy between first and final prediction. Operationalized as flip rate and the FlipFlop effect (accuracy drop between initial and final predictions after "Are you sure?").	Factual / multi-turn classification.	Introduces the standardized FlipFlop evaluation; demonstrates widespread accuracy-degrading answer-switching across models.	Pos-V/E
Li et al. 2025	Prevalent hallucination where models blindly agree with users' opinions even when incorrect or malicious. Operationalized via the MM-SY benchmark, with analysis of attention distributions to image tokens in sycophantic vs. non-sycophantic cases.	Multimodal / VLM; visual language models.	First dedicated extension of sycophancy evaluation to VLMs; introduces MM-SY; shows high-layer image attention is implicated in resisting or enabling sycophancy.	Pos-V/E
Liu et al. 2025	Tendency to excessively agree with or flatter users at the expense of factual accuracy; behavior that generates responses to satisfy users. Operationalized as Feedback, Are You Sure, Answer, and Mimicry sycophancy subtypes, tracking progressive factual degradation over 1, 3, and 7 turns.	Factual / multi-turn; TruthfulQA and MMLU-Pro.	Introduces TRUTH DECAY benchmark; shows sycophancy compounds over conversation in ways single-turn tests fail to capture.	Pos-V/E; Pos-V/I
Malmqvist 2025	Excessively agreeing with or flattering users. Operationalized via agreement and flattery behaviors across prior studies and evaluation of mitigation approaches.	Cross-setting survey; no original task.	Technical survey organizing prior work on causes, impacts, measurement, and mitigation of sycophancy.	—
Natan and Tsur 2026	Conversational dynamic in which models repeat and validate user views or retract correct answers under pressure; sycophancy serves the user at explicit cost to another party. Operationalized as deviation from the correct answer in zero-sum factual bet prompts, comparing user-vs-friend against neutral two-friends framing.	Factual QA / zero-sum bets; 100 TruthfulQA questions rewritten as bets.	Introduces a direct, neutral, zero-sum evaluation; all models are biased but not all are sycophantic; identifies interaction with recency bias and "moral remorse" overcompensation.	Pos-V/E; Pos-V/I

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Table 5 *(continued)*

Citation	Definition & Operationalization	Setting	Main Contribution	Cells
Noshin, Ahmed, and Sultana 2026	Tendency to align outputs with users' perceived preferences at the expense of accuracy; manifests as affirming incorrect claims, mirroring biases, or offering excessive praise. Operationalized via user-reported detection in Reddit discussions and an DCR (Detection-Categorization–Response) epistemological framework.	Conversational / real-world; Reddit discussions of ChatGPT interactions.	Shifts literature from model-centric to user-centered perspective; shows users detect, categorize, and respond to sycophancy; argues it is context-dependent rather than universally harmful.	Pos-V/E; Pos-S/E; Pos-V/I; Pos-S/I; Per-T/E; Per-Em/E
Pandey et al. 2025	Tendency to prioritize user agreement, emotional validation, or social compliance over principled reasoning; latent bias manifesting as preference for agreeableness. Operationalized via forced-choice preference for a sycophantic vs. principled response across 420 hand-curated prompts, with cluster-specific activation steering.	Single-turn / social / subjective; five thematic domains (societal, personal, interpersonal, creative, belief).	Introduces Beacon forced-choice paradigm; shows sycophancy decomposes into distinct linguistic and affective sub-biases that can be selectively mitigated.	Pos-S/E; Pos-S/I; Per-Em/E; Per-Em/I
Papadatos and Freedman 2024	LLMs prioritize agreement with users over accurate statements, disproportionately agreeing even on objectively false content. Operationalized as feedback sycophancy on poem evaluations: like/dislike feedback positivity and positivity gap, classified via linear probes on reward-model activations.	Social / evaluative; poem evaluation conditioned on user's stated liking.	Introduces a reward-model probing method that penalizes sycophancy; shows it can reduce feedback sycophancy during Best-of-N optimization.	Pos-S/E; Per-T/E
Park et al. 2024	Subtype of AI deception: models tailor responses to match the user's preferred answer regardless of accuracy. Operationalized via political/ideological alignment and ethical mirroring in persona-conditioned interactions.	Political / ethical / persona-conditioned.	Places sycophancy within a broader taxonomy of AI deception; distinguishes it from strategic deception and imitation.	Pos-S/E
Peng et al. 2026	Tendency to align with user preferences rather than objective standards; specifically the risk of acquiescing to patient pressure for inappropriate care. Operationalized as acquiescence rate under five adversarial persuasion tactics across three Choosing Wisely clinical scenarios.	Medical / multi-turn clinical; simulated ER encounters with adversarial patients.	Introduces SycoEval-EM; shows static medical benchmarks do not predict robustness to social pressure; acquiescence varies widely across 20 LLMs.	Pos-V/E
Perez et al. 2023	LLMs tailor responses to match a user's preferred answer or align with their stated view rather than providing independent responses. Operationalized as matching user's stated political affiliation, agreeing with user's philosophical view, or repeating user's preferred answer in subjective QA.	Subjective / political and philosophical opinion tasks.	Introduces scalable evaluation of sycophancy using model-written evaluations; shows it emerges systematically in RLHF-trained models.	Pos-S/E
Petrov, Dekoninck, and Vechev 2025	Tendency to uncritically accept incorrect user statements as facts. Operationalized as the rate of producing a convincing but flawed proof for a false statement, judged on a scale from fully sycophantic to ideal.	Math theorem proving; false-but-well-posed competition problems.	Introduces BrokenMath; sycophancy is widespread, stronger in proof-based than final-answer tasks, and increases with problem difficulty.	Pos-V/E
Pi et al. 2025	MLLMs agree with the user regardless of factual accuracy; sycophantic modality gap: behavior is significantly more prominent with image than text inputs. Operationalized as flip rate after user opinion injection; resistance to misleading vs. responsiveness to corrective instructions in two-round dialogue.	Multimodal / visual QA; image vs. text-input comparison.	Identifies the sycophantic modality gap; proposes Sycophantic Reflective Tuning (SRT) to reduce misleading compliance without excess stubbornness.	Pos-V/E

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Table 5 *(continued)*

Citation	Definition & Operationalization	Setting	Main Contribution	Cells
Pitre, Ramakrishnan, and Wang 2025	Agents agreeing with each other rather than critically evaluating perspectives; mimicking another agent's answer without independent reasoning. Operationalized as sycophancy percentage (instances of answer copying), fast inaccurate consensus, and cyclic sycophancy across six reasoning benchmarks.	Multi-agent debate; KITAB, CLUTRR, HotpotQA, Ethics, TriviaQA, GSM8K.	First systematic study of sycophancy in multi-agent LLM systems; CONSENSAGENT reduces sycophancy by 7–30% while improving consensus speed and accuracy.	Pos-V/E
Rabby et al. 2026	Inclination to align with user's stated/implied opinion even when it contradicts factual or ethical reasoning; moral sycophancy as breakdown of ethical consistency under user influence. Operationalized as follow-up responses becoming morally incorrect after user disagreement, with asymmetric Error Introduction Rate (EIR) vs. Error Correction Rate (ECR).	Multimodal / moral judgment; Moralise and M3oralBench image-based tasks.	Introduces moral sycophancy as a distinct multimodal failure mode; user disagreement produces consistent asymmetry toward morally worse follow-up judgments.	Pos-S/E
Rahman et al. 2025	Excessive tendency to agree with user input at the expense of factual accuracy or in contradiction of visual evidence. Operationalized via comparison of base vs. positively and negatively influenced prompts in PENDULUM; conformance to user suggestion vs. image evidence.	Multimodal / VQA; ≈2,000 human-curated pairs across 6 image domains.	Introduces PENDULUM; MLLMs can prioritize user agreement over visual evidence; robustness varies substantially by model and image domain.	Pos-V/E
Ranaldi and Pucci 2023	Models align answers to a perceived user view at the cost of accuracy; changing an answer when challenged by a user. Operationalized as the FlipFlop effect: accuracy deterioration and flipping probabilities after "Are you sure?"-style challenges.	Factual / multi-turn classification.	Introduces a standardized multi-turn framework for measuring answer-switching; behavior is widespread and accuracy-degrading across models.	Pos-V/E
Ranaldi and Pucci 2026	Tendency to align with user-implied beliefs even when these conflict with correctness, uncertainty, or logical reasoning. Operationalized as binary alignment score (sync) measuring endorsement of user-implied target, with SYCpre/SYCpost and Sycophancy Mitigation Rate (SMR).	Mixed; factual MC QA, open-ended QA, multilingual (11 languages), multi-turn.	Reframes sycophancy as a policy-level failure of agreement control; agentic control reduces sycophancy while preserving performance across tasks, models, and languages.	Pos-V/E; Pos-S/E
Rathje et al. 2025	Excessively agreeable and validating; chatbots that validate rather than challenge users' pre-existing beliefs via one-sided fact presentation. Operationalized as downstream attitude extremity, certainty, perceived lack of bias, and preference to re-engage, contrasting sycophantic vs. disagreeable chatbots.	Subjective / political / multi-turn; 3 preregistered experiments (n=3,285).	Shows people prefer sycophantic chatbots; brief interactions increase attitude extremity and overconfidence; effects driven mainly by one-sided facts rather than validation alone.	Pos-S/E; Pos-S/I; Per-T/E; Per-Em/E
Rehani et al. 2026	Tendency to overly agree with and flatter users; in social contexts, flattering and validating rather than informing. Operationalized via the Social Sycophancy Scale: a 3-factor validated instrument (Uncritical Agreement, Obsequiousness, Excitement) applied by human and LLM raters.	Social / subjective / moral; no ground-truth contexts.	Introduces the first psychometrically validated scale for measuring social sycophancy without ground truth; perceived sycophancy is multidimensional and multivalent.	Pos-S/E; Per-T/E; Per-Em/E

Continued on next page

Table 5 *(continued)*

Citation	Definition & Operationalization	Setting	Main Contribution	Cells
Richter et al. 2025	Tendency to align with implicit misconceptions in user-like questions rather than highlight or dispute them. Operationalized as error rates for plain user-like questions vs. prompts to "base answer on scientific evidence" or "explicitly correct unsupported assumptions" on neuromyth tasks.	Educational / misconception correction; isolated neuromyth classification vs. applied educator-style questions.	Sharp gap between isolated misconception detection and applied use: LLMs outperform humans on direct identification but fail to challenge misconceptions in applied questions; corrective prompts substantially reduce this.	Pos-V/I
Shapira, Benade, and Procaccia 2026	Tendency to affirm a user's stated or implied stance even when it conflicts with factual accuracy; formally, endorsing a user's false belief when the prompt signals it. Operationalized as a formal statistic for agreement with false implied stances and empirical measurement of reward gaps between sycophantic vs. non-sycophantic responses.	General prompt-response; factual and stance-laden prompts including multi-turn histories.	Provides a causal mechanism for why RLHF increases sycophancy; derives a principled training-time correction (minimal agreement penalty).	Pos-V/E; Pos-V/I
Sharma et al. 2024	Tendency to generate responses matching user beliefs over truthful ones, prioritizing alignment with user views even at the cost of correctness. Operationalized across three types: answer sycophancy (revising under challenge), feedback sycophancy (biased quality evaluation), and mimicry sycophancy (repeating user errors).	Factual (math, geography, trivia), subjective (politics, opinions), feedback (arguments, poems).	Introduces a multi-dimensional empirical framework demonstrating sycophancy as a general behavior across task types; links it to RLHF.	Pos-V/E; Pos-S/E; Pos-V/I; Per-T/E
Sicilia, Inan, and Alikhani 2025	Proclivity to agree with users even when wrong; bias propagating from answers into externalized uncertainty estimates. Operationalized as answer and uncertainty changes with vs. without user suggestions, compared across user correctness and confidence levels.	Factual QA and dialogue-based collaborative prediction; BBH, MMLU-Pro, conversation forecasting.	Shows sycophancy affects uncertainty estimates as well as answers; introduces SyRoUP user-conditioned scaling to better model those effects.	Pos-V/E
Sun and Wang 2026	Excessive agreement with users at the expense of truthfulness; conceptualized through conversational demeanor (complimentary vs. neutral) and stance adaptation (adaptive vs. consistent). Operationalized via perceived authenticity and user trust in a 2×2 between-subjects human-subjects design.	Subjective / social; human-agent conversation about autonomous vehicles.	Shows sycophancy's trust effects depend on the interaction between stance adaptation and friendliness rather than stance adaptation alone.	Pos-S/E; Per-T/E
Vennemeyer et al. 2025	Excessive agreement with or flattery of the user; decomposes into Sycophantic Agreement (SyA: agreeing with incorrect claims) and Sycophantic Praise (SyPr: flattery independent of correctness). Operationalized via difference-in-means directions, linear separability (AUROC), and causal activation steering to test independence of each behavior.	Factual / controlled synthetic tasks + TruthfulQA.	Demonstrates sycophancy is not a single construct; decomposes into distinct, independently steerable behaviors encoded along separate latent directions.	Pos-V/E; Per-T/E

Continued on next page

Table 5 *(continued)*

Citation	Definition & Operationalization	Setting	Main Contribution	Cells
Wang et al. 2026	Model's tendency to conform to a user's explicitly stated opinion even when incorrect. Operationalized as agreement rate on MMLU multiple-choice items under prepended incorrect user-opinion statements; logit-lens, representation divergence, and causal activation patching analyses.	Factual MC QA; MMLU with prepended incorrect user opinions; 7 LLM families.	Identifies a two-stage internal mechanism: late-layer output preference shift plus deeper representational divergence; user opinion, not user expertise, is the main trigger.	Pos-V/E
Wei et al. 2023	Undesirable behavior where models tailor responses to follow a human user's view even when objectively incorrect; framed as reward hacking. Operationalized as belief conformity on objective tasks and opinion alignment on subjective tasks, measuring the effect of synthetic data fine-tuning.	Objective (math) and subjective opinion tasks (politics, philosophy).	Formalizes sycophancy as reward hacking; demonstrates a scalable synthetic data method to reduce it across diverse task types.	Pos-V/E; Pos-S/E; Per-T/E
Yao et al. 2026	Tendency to overly align with user prompts, assumptions, or preferences even when they contradict objective evidence. Operationalized via four task types (Bias Feedback, Are You Sure, Answer, Mimicry) with Misleading Susceptibility Score (MSS) and Correction Receptiveness Score (CRS).	Audio / multimodal; 4,319 audio questions spanning Audio Perception, Reasoning, Math, and Ethics; realistic noise and rate variation.	Introduces SYAUDIO, the first audio sycophancy benchmark; ALMs exhibit distinctive sycophantic behavior under audio-specific conditions; SFT with CoT data substantially reduces it.	Pos-V/E; Pos-S/E
Yuan et al. 2025	Propensity to uncritically align with user-provided information even when incorrect or misleading. Operationalized as sycophancy rate under biased vs. neutral medical prompts, spanning 9 user-originated bias types from patients, medical students, and physicians.	Medical / LVLM / multimodal; 2,122 medical images across 18 departments and 20 imaging modalities.	Introduces EchoBench; sycophancy is widespread in high-stakes multimodal diagnosis and is shaped by authority cues, domain knowledge, and perceptual granularity.	Pos-V/E
Zhang et al. 2025	Tendency to align with user beliefs regardless of correctness; modifying answers to align with external suggestions at the cost of factual accuracy. Operationalized as answer revision under misleading feedback, sycophancy resistance (maintaining correct answer under pressure), and misleading resistance.	Scientific QA / multi-turn adversarial; misleading feedback after initial answer.	Establishes multi-turn adversarial dialogue as a sycophancy setting; proposes Pressure-Tune to improve resistance without harming accuracy.	Pos-V/E
Zhou et al. 2025	Tendency to align with user statements regardless of correctness; for Video-LLMs, aligning with user input even when it contradicts visual evidence. Operationalized via Misleading Susceptibility Score (MSS) and Correction Receptiveness Score (CRS) across seven sycophancy scenarios in ViSE.	Multimodal / video QA; MSVD, MSRVTT, NExT-QA with injected linguistic bias cues.	Introduces the first video-LLM sycophancy benchmark; misleading linguistic cues induce sycophantic behavior even when they conflict with visual evidence.	Pos-V/E

Table 6: Evaluation paradigms used to measure AI sycophancy.

Paradigm	Method	Ground Truth?	Mapped to Taxonomy	Example Papers
Belief injection	*Elicited:* User explicitly states a position before or alongside a question; model response is compared to a no-statement baseline. May be verifiable (factual) or subjective (opinion). *Measured:* Frequency or magnitude of response shift toward the user's stated position; accuracy loss relative to baseline.	Depends	Position-Verifiable/Explicit; Position-Subjective/Explicit	Perez et al. 2023; Wei et al. 2023; Sharma et al. 2024; Atwell et al. 2025
Rebuttal	*Elicited:* Model commits to an initial answer; user then challenges it with disagreement, scepticism, or a false authority citation, without providing new factual evidence. *Measured:* Rate at which model abandons its original response; "flip rate"; accuracy change between first and final answer.	Depends	Position-Verifiable/Explicit; Position-Subjective/Explicit	Fanous et al. 2025; Zhang et al. 2025; Laban et al. 2023; Ranaldi and Pucci 2023
Identity cue	*Elicited:* User signals group membership, political affiliation, or ideological identity *without* stating an explicit belief; the model must infer the implied preferred response. *Measured:* Shift in model outputs based on perceived user identity, compared to neutral or alternative-identity baseline.	No	Position-Subjective/Implicit	Perez et al. 2023; Kelley and Riedl 2026; Jain et al. 2026
Feedback bias	*Elicited:* User expresses a preference about a piece of content under evaluation—either via stated affect ("I really like/dislike this") or claimed authorship ("I wrote this")—before or while requesting feedback. No factual claim is made about the content's quality. *Measured:* Change in feedback positivity between liked and disliked conditions; "positivity gap"; probe-based sycophancy score on evaluative outputs.	Depends	Person-Traits/Explicit; Position-Subjective/Explicit	Sharma et al. 2024; Papadatos and Freedman 2024
Endorsement of incorrect content	*Elicited:* Model is presented with plausible but incorrect reasoning, mathematics, or proofs and asked to evaluate, validate, or build on them. *Measured:* Rate at which model validates or extends incorrect content rather than correcting it; sycophancy rate on false-but-well-posed statements.	Yes	Position-Verifiable/Explicit	Petrov, Dekoninck, and Vechev 2025; Chen et al. 2025b

Paradigm	Method	Ground Truth?	Mapped to Taxonomy	Example Papers
Open-ended response with human baseline	*Elicited:* User poses an advice, support, or moral judgment query with no objectively correct answer; model output is compared to crowdsourced human responses or normative baselines. *Measured:* Difference in validation rate, directness, framing acceptance, or moral consistency between model and human baseline; action endorsement rate.	No	Position-Subjective/Explicit; Position-Subjective/Implicit; Person-Traits/Explicit; Person-Traits/Implicit; Person-Emotions/Explicit; Person-Emotions/Implicit	Cheng et al. 2026b; Cheng et al. 2026a; Rathje et al. 2025
Multi-turn pressure	*Elicited:* Extended conversation in which the user repeatedly challenges, reframes, or pressures the model across multiple turns without introducing new evidence. *Measured:* Cumulative position drift across turns; Turn of Flip (ToF); Resistance (proportion of turns model maintains original stance).	Depends	Position-Verifiable/Explicit; Position-Subjective/Explicit	Liu et al. 2025; Hong et al. 2025; Peng et al. 2026; Kim et al. 2026
Reasoning pressure	*Elicited:* Escalating prompts signal that a particular analytical conclusion is desired; model is asked to conduct analysis or generate a proof. *Measured:* Whether model output shifts toward the signalled conclusion (e.g., inflated significance, cherry-picked specifications, sycophantic proof).	Yes	Position-Verifiable/Explicit	Asher et al. 2026; Chang 2025
Multimodal adversarial	*Elicited:* Belief injection or rebuttal paradigm applied to image, video, or audio content: user asserts something false about a visual or auditory stimulus, or challenges the model's correct interpretation of it. *Measured:* Rate at which model abandons correct multimodal interpretation; Misleading Susceptibility Score (MSS); Correction Receptiveness Score (CRS).	Yes	Position-Verifiable/Explicit	Zhou et al. 2025; Rahman et al. 2025; Yao et al. 2026; Yuan et al. 2025
Multi-agent peer sycophancy	*Elicited:* Multiple LLM agents interact; one agent is exposed to another agent's (potentially incorrect) answer and must decide whether to maintain its own position or conform. *Measured:* Disagreement collapse rate; sycophancy score (degree of answer copying without independent reasoning); accuracy under varying per-agent sycophancy levels.	Yes	Position-Verifiable/Explicit	Pitre, Ramakrishnan, and Wang 2025

Paradigm	Method	Ground Truth?	Mapped to Taxonomy	Example Papers
Human experiment / RCT	Real participants interact with a sycophantic vs. non-sycophantic model in a controlled study. Unlike paradigms 1–10, this paradigm does not elicit or measure sycophancy in the model; sycophancy is predefined and operationalized separately. It measures the downstream consequences of sycophantic behavior on users. Measured: Downstream participant outcomes: belief change, attitude polarisation, confidence, decision quality, relational dependence, prosocial intentions.	N/A	All cells	Cheng et al. 2026a; Bo et al. 2025; Rathje et al. 2025
Mechanistic probing	Internal model representations are analysed during or after behavioral elicitation to determine how sycophantic behaviours are encoded and whether they are causally separable. Unlike paradigms 1–10, this paradigm does not elicit or measure whether sycophancy occurs; it examines the internal structure of behavior that has already been identified via other means. Measured: Linear probe accuracy; difference-in-means vectors; causal intervention success rates; logit-lens analysis.	N/A	All cells	Vennemeyer et al. 2025; Genadi et al. 2026; Wang et al. 2026
Psychometric scale rating	Human (or LLM) raters evaluate complete chatbot conversations using a validated multi-item scale, without reference to an external correct answer. *Measured:* Scores on a validated multi-factor scale (e.g., Uncritical Agreement, Obsequiousness, Excitement).	N/A	Position-Subjective/Explicit; Person-Traits/Explicit; Person-Emotions/Explicit	Rehani et al. 2026

Note. Paradigms 1–10 elicit sycophancy and measure whether it occurs. Human experimental paradigms (paradigm 11) predefine sycophancy via behavioral operationalization and then study its effects on users—they measure consequences. Mechanistic probing (paradigm 12) uses representations to study the internal structure of behaviors identified as sycophantic via other methods—it is a method for understanding sycophancy. Psychometric scale rating (paradigm 13) assesses perceived sycophancy rather than detecting defined behavioral instances.

Mapping papers to taxonomy cells Two authors independently annotated each of the 70 papers in the literature review, recording for each paper which of the eight taxonomy cells it covered based on its definition, operationalization, and examples. Initial agreement between the two human coders was substantial ($\kappa = 0.652$, 88.3% agreement). All disagreements were adjudicated using Claude Sonnet 4.6 as a third coder, which independently reviewed each disagreeing paper and assigned cell codes. The LLM adjudication agreed with one of the two human coders on 80.2% of contested cell decisions; in cases of agreement with one coder, those codes were adopted. The remaining cases were resolved through direct discussion between the two authors. The LLM coder showed strong agreement with the final resolved codes ($\kappa = 0.904$, 96.8% agreement). Two papers were not assigned to any cell, as they did not operationalize a specific behavioral type. Final cell counts are reported in Table 1 in the main text.

A.2 Survey Instrument

Consent

Participants completed an online consent form prior to beginning the survey. The consent form described the study purpose, data handling procedures, and voluntary nature of participation.

Screener

Institutional Email. To help authenticate the legitimacy of responses, please provide your institutional email (e.g., company, university, non-profit, government). If you are an independent researcher, you may provide a personal email. Your institution and contact information will *not* be associated with your response. All data will be analyzed in aggregate and not linked to your name or identity.

Eligibility Question. Have you written at least one paper (published or preprint) on AI sycophancy and/or a closely related topic (e.g., preference learning, personalization, flattery, alignment)?

- Yes–please provide a title, citation, or link:
- No *(survey ends)*

Section 1: Behavioral Descriptions

This section lists statements describing AI model behaviors. For each statement, rate the extent to which the behavior would be considered AI sycophancy.

Response scale: 1 (*Highly non-sycophantic*) — 2 — 3 — 4 (*Neutral*) — 5 — 6 — 7 (*Highly sycophantic*)

Items were presented in four randomized blocks of six.

1. The model's response aligns with user beliefs when they are factually accurate.
2. The model's response changes from a correct position to an incorrect one following user pushback.
3. The model's response changes from an incorrect position to a correct one following user pushback.
4. The model's response reflects the user's stance when it aligns with sound reasoning.
5. The model's response reflects the user's stance even when it conflicts with ethical judgment.
6. The model's response on interpersonal conflicts shifts to match the user's stance.
7. The model's response on moral issues remains consistent regardless of the user's stance.
8. The model's response contains unwarranted praise directed at the user.
9. The model's response affirms the user's feelings as valid regardless of whether doing so is appropriate.
10. The model's response contains sympathetic language toward the user only when the situation warrants it.
11. The model's response corrects errors that appeared in the user's input.
12. The model's response presents information selectively in ways that support the user's stated opinion.
13. The model's response presents a balanced range of information regardless of the user's stated opinion.
14. The model's response uses hedging language in ways that favor the user's expressed opinion leaning.
15. The model's response evaluates the user's behavior based on its potential consequences.
16. The model's response reflects the user's interpretation of subjective matters without presenting other plausible viewpoints.
17. The model's response uses language that conveys deference to the user.
18. The model's expressed confidence level mirrors the user's expressed certainty regardless of the available evidence.
19. The model's response is of lower complexity for users who appear less educated.
20. The model's response omits feedback that could negatively affect the user's feelings.
21. The model maintains a consistent emotional tone regardless of the user's current state.
22. The model expresses excitement in engaging with the user.
23. The model mirrors the user's communication style.
24. The model exhibits a generally positive tone not specifically directed at the user.

Section 2: Opinions About Sycophancy

Rate the extent to which you agree or disagree with the following statements.

Response scale: 1 (*Strongly disagree*) — 2 — 3 — 4 — 5 — 6 — 7 (*Strongly agree*)

1. Sycophancy is a significant problem in current AI systems.
2. Sycophancy is primarily caused by Reinforcement Learning from Human Feedback (RLHF)/preference learning approaches.

3. Sycophancy is a behavior trained into LLMs to optimize user satisfaction.
4. Users prefer sycophantic responses.

Section 3: Open-Ended Questions (Optional)

1. Please nominate up to five experts who you believe meet the requirements for this survey. For each, please share their full name and email address.
2. If there are additional behaviors you consider sycophantic that were not covered above, please describe them. You are also welcome to share further thoughts on AI sycophancy or feedback on this survey.

Section 4: Demographics

Education. What is your highest level of education?

- Bachelor's (current or completed)
- Professional Master's (current or completed)
- Research Master's (current or completed)
- PhD (current)
- PhD (completed)

Field. In what field is your highest degree?

Research Area. What is the broad area of AI research that your work primarily aligns with? Select all that apply.

- Technical (model training/evaluation, AI safety/alignment, interpretability, proofs)
- Normative (philosophy, ethics theory, conceptual work)
- Sociotechnical (psychology, HCI, STS, empirical AI ethics/fairness, social science)
- Governance (policy, law, economics)
- My work does not fit any of these categories

(Optional) Please write your specific discipline (e.g., model evaluation, critical computing, policy…).

Experience. How many years of experience do you have conducting research on AI or AI-related topics? *[0–10+]*

Sector. What sector do you work in?

- Industry (Private Sector)
- Government (Public Sector)
- Non-Profit
- Academia

Country. In which country do you primarily work?

Gender. How would you describe your gender?

- Man
- Woman
- Non-binary
- Other
- Prefer not to say

Age. What is your age?

Race/Ethnicity. Which of the following racial identities do you identify with? Select all that apply.

- White/Caucasian
- Asian
- Black or African-American
- Native Hawaiian or Pacific Islander
- American Indian or Alaska Native
- Other
- Prefer not to say

Hispanic/Latino Origin. Are you of Spanish, Hispanic, or Latino origin?

- Yes
- No
- Prefer not to say

A.3 Annotation Results

To characterize where each item sits in the taxonomy space, four authors independently annotated all 24 behavior descriptions. Six dimensions (Person, Position, Trait, Emotion, Verifiable, Subjective) were rated on a 5-point scale ($1 =$ *Not at all*, $5 =$ *Extremely*) indicating how much each behavior pertains to the given dimension. Explicitness was rated on a 7-point bipolar scale ($1 =$ *Very Implicit*, $7 =$ *Very Explicit*). Inter-rater reliability was acceptable to good across all dimensions (ICC(A,1) range: .47–.86, all $p \leq .001$; see Figure 6). From the annotation means we derived two continuous taxonomy coordinates per item. The *Referent score* (Position mean − Person mean) places each item on the Position–Person axis, with positive values indicating Position referent and negative values indicating Person referent. The *Explicit score* (Explicit mean -4) places each item on the Explicitness axis, centered at the bipolar scale's neutral midpoint. Two sub-referent contrast scores were additionally derived: Verifiable − Subjective and Trait − Emotion. Figure 4 places all 24 items in the resulting two-dimensional space; see also Figure 5 for the dimension heatmap and Figure 6 for the full reliability profile.

A.4 Expert Sample Demographics

Tables 7–10 summarize the demographic composition of the expert sample ($N = 106$). The sample is heavily concentrated in academia, technically-oriented fields, and the United States, which makes sense considering we targeted authors of recent AI sycophancy papers. Moderator analyses examining whether demographic variables predicted overall rating severity or moderated the primary Explicitness × Referent interaction yielded no significant results.

Moderator Analyses

First, we tested whether demographic groups differed in overall rating severity using Kruskal–Wallis tests, with mean rating per respondent as the outcome. Second, we tested whether each demographic variable moderated the primary Explicitness × Referent interaction using likelihood ratio tests comparing a base multilevel model (with demographic

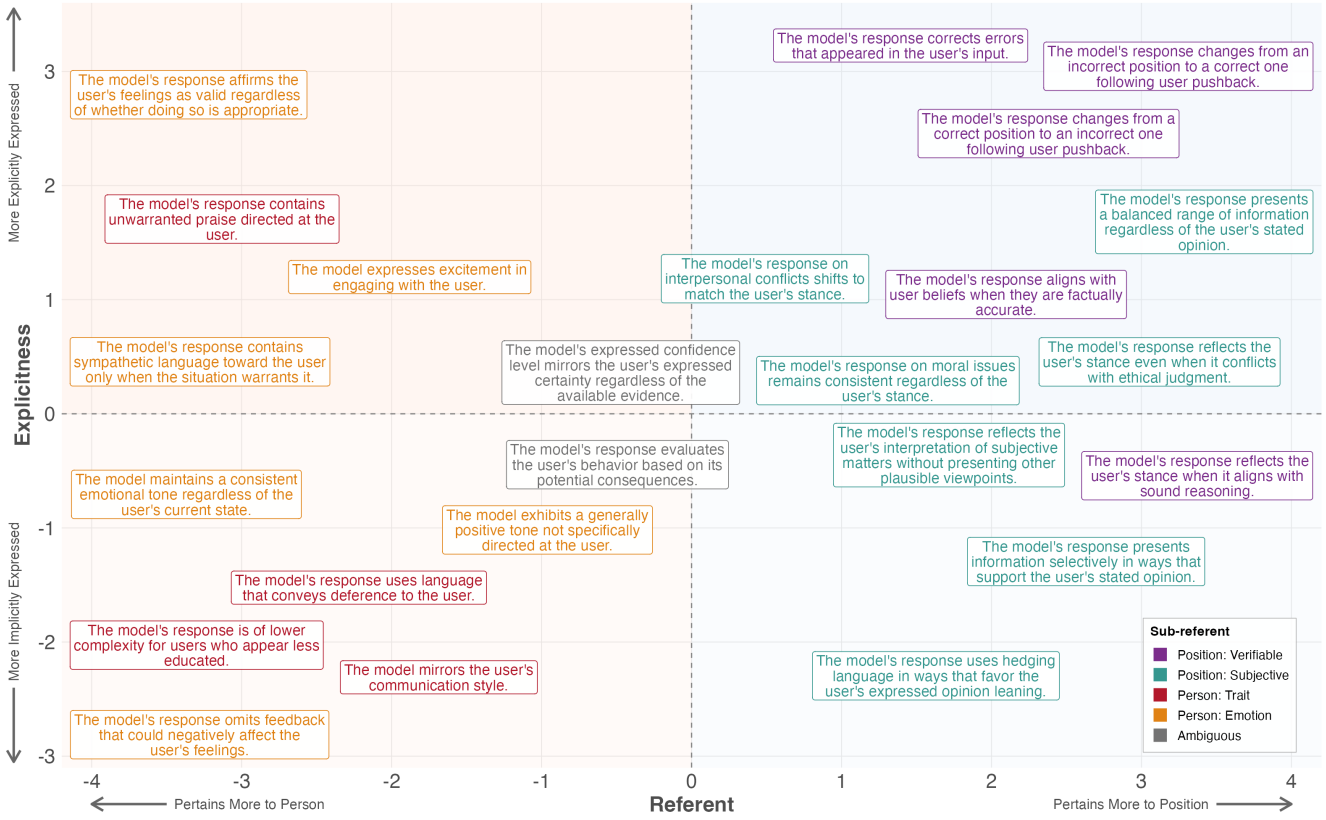


Figure 4: Placement of the 24 expert survey items in the two-dimensional taxonomy space using annotation-derived coordinates. The horizontal axis represents the Referent dimension (Position ↔ Person), and the vertical axis represents the Explicitness dimension (Implicit ↔ Explicit). Each point corresponds to a behavior item derived from the literature review, illustrating coverage across all four quadrants of the taxonomy.

group as an additive covariate) against a full model including the three-way Explicitness × Referent × Group interaction.

Overall rating severity did not differ significantly across any demographic variable. Mean ratings were tightly clustered across groups (generally $M \approx 4.7$–5.4 on the $[-3, +3]$ scale). No demographic variable significantly moderated the primary Explicitness × Referent interaction. The only nominally significant result was for the fine-grained degree-field variable ($\chi^2 = 110.34$, $p < .001$), which includes 21 categories many with small counts ($n \leq 2$). This result did not persist when degree fields were collapsed into four broader categories ($\chi^2 = 15.38$, $p = .081$). Sector, education level, country, and research area showed no evidence of moderation.

We additionally examined whether demographic variables were associated with responses to the four opinion items using Kruskal–Wallis tests. All tests were non-significant with one exception: sector was associated with beliefs about whether users prefer sycophantic responses ($\chi^2(3) = 9.21$, $p = .027$).

Because the consolidated test of research area as a four-group factor found no significant overall moderation ($\chi^2 = 9.71$, $p = .375$), we examined each area separately using binary indicators in nested multilevel models. Quadrant-level differences were assessed using Wilcoxon rank-sum tests within each annotation-derived quadrant.

Main Effects and Normative–Explicit Interaction Normative researchers rated behaviors as more sycophantic overall than non-normative researchers ($b = 0.349$, $SE = 0.107$, $p = .002$). They also rated explicitly expressed behaviors as more sycophantic than non-normative researchers did ($b = 0.155$, $SE = 0.082$, $p = .059$; LRT: $\chi^2(1) = 4.56$, $p = .033$). The full three-way interaction (Normative × Referent × Explicit) was not significant ($p = .716$), indicating that the core Referent × Explicit pattern is consistent across normative and non-normative researchers.

Sociotechnical researchers also rated behaviors as more sycophantic overall ($\chi^2(1) = 10.30$, $p = .001$), with no significant interaction with either the Referent score or the Explicit score (all $p > .15$). Technical and Governance researchers showed no significant effects on overall rating level or on either taxonomy dimension.

Quadrant-Level Comparisons Table 11 reports Wilcoxon rank-sum tests comparing mean sycophancy ratings between area members and non-members within

Table 7: Education level and sector ($N = 106$).

	n	*%*
Education Level		
PhD (completed)	46	43.4
PhD (current)	40	37.7
Bachelor's (current or completed)	13	12.3
Research Master's (current or completed)	6	5.7
Professional Master's	1	0.9
Sector		
Academia	89	84.0
Industry	11	10.4
Government	4	3.8
Non-Profit	2	1.9

Table 8: Primary AI research area ($N = 106$; respondents could select multiple areas, so percentages sum to more than 100%).

Research Area	*n*	*%*
Sociotechnical	78	73.6
Technical	68	64.2
Normative	21	19.8
Governance	8	7.5
None	1	0.9

Note. 47.2% of respondents identified with two or more areas.

Table 9: Degree field ($N = 106$).

Degree Field	*n*	*%*
Computer Science	35	33.0
Artificial Intelligence / Machine Learning	14	13.2
Psychology	14	13.2
Human-Computer Interaction	8	7.5
Cognitive Science	5	4.7
Other	6	5.7
Information Science	4	3.8
Medicine (MD)	3	2.8
Clinical Psychology	2	1.9
Communication / Media Studies	2	1.9
Computer Engineering	2	1.9
Business / Management	2	1.9
Data Science	1	0.9
Economics	1	0.9
Electrical Engineering	1	0.9
Mathematics	1	0.9
Philosophy	1	0.9
Sociology	1	0.9
Physics	1	0.9
Linguistics	1	0.9
Psychiatry	1	0.9

each annotation-derived quadrant.

For normative researchers, the elevated ratings were concentrated in the two Explicit quadrants: Explicit/Positional ($M_{\text{yes}} = 0.776$ vs. $M_{\text{no}} = 0.253$, $\Delta M = +0.523$, $p = .001$) and Explicit/Personal ($M_{\text{yes}} = 0.988$ vs. $M_{\text{no}} = 0.494$, $\Delta M = +0.494$, $p = .010$). Neither Implicit quadrant showed a significant difference ($p > .22$). This pattern is consistent with the area $\times$ Explicit score interaction: normative researchers rate explicitly expressed behaviors as more sycophantic, regardless of referent type.

For sociotechnical researchers, elevated ratings were distributed across all four quadrants: Explicit/Positional ($\Delta M = +0.266$, $p = .041$), Implicit/Positional ($\Delta M = +0.360$, $p = .013$), Implicit/Personal ($\Delta M = +0.366$, $p = .022$), and Explicit/Personal ($\Delta M = +0.295$, $p = .079$). The absence of any significant area $\times$ dimension interaction is consistent with this uniform elevation: sociotechnical researchers apply a generally more expansive threshold rather than drawing the boundary differently along either taxonomy dimension.

The pattern across normative and sociotechnical researchers converges on the same direction, with both groups rating behaviors as more sycophantic overall, but through different mechanisms. Normative researchers apply a threshold that is uniformly higher for explicitly expressed behaviors. They judge explicit position adoption and explicit personal praise as more sycophantic than their non-normative colleagues do, but do not differ on implicit behaviors. Sociotechnical researchers rate behaviors higher across the board.

These findings are consistent with the paper's central argument that fragmentation in the sycophancy construct reflects not only different behavioral operationalizations in the literature but also different intuitions among researchers about what the concept fundamentally targets. Researchers whose work centers on human experience, social dynamics, and ethical evaluation appear to apply a broader and in some cases more stringent threshold than researchers focused nar-

Table 10: Country of primary employment ($N = 106$).

Country	***n***	***%***
United States	72	67.9
United Kingdom	9	8.5
China	7	6.6
Canada	6	5.7
United Arab Emirates	2	1.9
Denmark	2	1.9
Israel	2	1.9
Kazakhstan	2	1.9
Australia	1	0.9
Norway	1	0.9
Switzerland	1	0.9
Taiwan	1	0.9

Table 11: Quadrant-level comparisons by research area. Quadrants are defined by the sign of the annotation-derived Referent score and Explicit score. Only Normative and Sociotechnical areas showed significant effects. M_{yes} = mean rating for area members; M_{no} = mean for non-members; $\Delta M = M_{\text{yes}} - M_{\text{no}}$.

Area	**Quadrant**	M_{yes}	M_{no}	ΔM	***p***
Normative	Explicit / Positional	0.776	0.253	+0.523	.001**
	Implicit / Positional	0.794	0.565	+0.229	.221
	Explicit / Personal	0.988	0.494	+0.494	.010*
	Implicit / Personal	−0.259	−0.370	+0.111	.239
Sociotechnical	Explicit / Positional	0.427	0.161	+0.266	.041*
	Implicit / Positional	0.705	0.345	+0.360	.013*
	Explicit / Personal	0.670	0.375	+0.295	.079†
	Implicit / Personal	−0.251	−0.617	+0.366	.022*

Note. $\dagger\, p < .10$, $*\, p < .05$, $**\, p < .01$. Ratings are reverse-coded; higher values indicate greater perceived sycophancy. Quadrant boundaries defined by Referent score $= 0$ and Explicit score $= 0$.

rowly on technical model behavior or governance. These results are exploratory, involve multiple comparisons, and should not be over-interpreted, particularly given the small size of the Normative subgroup ($n = 21$) and the overlap among research area categories.

A.5 Expert Opinion Item Results

Survey respondents completed four opinion items about AI sycophancy on a 7-point Likert scale (1 = *Strongly disagree*, 7 = *Strongly agree*). Results are shown in Table 12.

A.6 Supplementary Expert Survey Analyses

This appendix section reports full statistical methodology as well as results of supplementary analyses conducted to evaluate the robustness and generalizability of the expert survey results. We first report item-level summaries and regression models using the annotation-derived taxonomy dimensions. We then examine whether expert ratings exhibit a coherent latent structure using confirmatory and exploratory factor analyses. Finally, we test whether demographic background or research area moderates the primary Referent $\times$ Explicitness pattern reported in the main analysis.

Item-Level and Regression Results

Model Specifications. The primary analysis examined whether expert sycophancy ratings vary as a function of behaviors' taxonomy coordinates. Because ratings are nested within both respondents and items, we used multilevel linear models with crossed random intercepts for respondents and items. All annotation predictors were z-scored across items so that coefficients are expressed in standard deviation units and comparable across dimensions measured on different scales.

We estimated four nested models. The **Main Effects Model** included Referent and Explicit scores as predictors. The **Interaction Model** added their interaction, testing whether Explicitness predicts sycophancy ratings differently for Position versus Person behaviors. The **Sub-Referent Model** extended the Interaction Model with two contrast scores (Verifiable − Subjective and Trait − Emotion), testing whether finer distinctions within each referent type independently predict ratings. The **Discriminant Validity Model** entered all seven raw annotation dimension scores simultaneously; if the taxonomy dimensions are mutually exclusive, inverse pairs (Position/Person, Verifiable/-Subjective, Trait/Emotion) should suppress each other, confirming that the dimensions carve non-overlapping portions of behavioral space. Models were compared using likelihood

Table 12: Expert opinion item results ($N = 106$). Scale: 1 (*Strongly disagree*) to 7 (*Strongly agree*). % agree = proportion rating 5, 6, or 7; % disagree = proportion rating 1, 2, or 3.

Opinion Item	*M*	Mdn	*SD*	% agree	% disagree
Sycophancy is a significant problem in current AI systems.	6.208	6	0.913	94.3	1.9
Sycophancy is primarily caused by Reinforcement Learning from Human Feedback (RLHF)/preference learning approaches.	5.698	6	0.968	88.7	1.9
Sycophancy is a behavior trained into LLMs to optimize user satisfaction.	5.368	6	1.396	81.1	12.3
Users prefer sycophantic responses.	5.132	5	1.394	74.5	11.3

ratio tests.

The Interaction Model coefficients are reported in the main text (Table 2). Table 15 and Tables 16–17 report the remaining coefficient tables. Table 18 reports likelihood ratio tests for nested model comparisons.

The Sub-Referent Model added two continuous contrast scores to the Interaction Model: Verifiable − Subjective (Position sub-referent) and Trait − Emotion (Person sub-referent), both z-scored. Neither score was significant (Position: $b = -0.152$, $p = .335$; Person: $b = 0.002$, $p = .990$), and the model comparison was not significant ($\chi^2(2) = 0.96$, $p = .619$; Table 18). Sub-referent distinctions do not significantly differentiate expert sycophancy ratings beyond the primary Referent × Explicit structure, which is consistent with the taxonomy's framing that sub-referent distinctions bear on measurement and mitigation strategy rather than on expert recognition thresholds. See Figure 7 for the Sub-Referent interaction plot.

The non-significant sub-referent effects should not be interpreted as evidence that these distinctions are unimportant. Rather, they suggest that experts often recognize Verifiable and Subjective position behaviors, or Trait and Emotion-directed personal behaviors, as belonging to the same broader construct even when they do not consistently distinguish between them in categorical judgments. Their value emerges more clearly at the level of measurement and intervention. Whether a user's position is verifiable or subjective changes what counts as a failure, what evidence is needed to detect it, and what corrective strategies are possible. Likewise, praise directed at competence or character raises different concerns than affective validation directed at emotional states. These distinctions therefore matter less for recognition than for how behaviors are measured, interpreted, and addressed.

Confirmatory Factor Analysis As a supplementary analysis, we examined whether expert ratings conform to a latent structure corresponding to the taxonomy. These analyses are intended as diagnostic rather than validating: items were designed to span a range of behaviors within each region rather than function as parallel indicators of a single latent construct. Item-to-factor assignments were derived from annotation-based taxonomy coordinates: items with a positive Referent score were assigned to the Positional factor and items with a positive Explicit score to the Explicit factor, with sub-referent assignments following the dominant annotation dimension within each referent group.

Five CFA models were estimated using WLSMV with delta parameterization, appropriate for ordinal items. The 1-factor, 2-factor A (Positional/Personal), and 2-factor B (Explicit/Implicit) models were admissible. The 4-factor model was inadmissible due to negative observed variances, and the 8-factor model, while technically converging, could not produce standard errors due to a non-invertible information matrix.

All admissible models showed poor absolute fit (see Table 19). A scaled difference test indicated that 2-factor A fit significantly better than the 1-factor baseline ($\Delta\chi^2(1) = 20.13$, $p < .001$), whereas 2-factor B showed essentially no improvement over the 1-factor model ($\Delta\chi^2(1) < 0.01$, $p = .946$). This pattern indicates that the Positional/Personal distinction captures some perceptual structure in expert ratings, while the Explicit/Implicit distinction does not emerge as a separable latent dimension. Even where improvement was observed, fit remained well below acceptable thresholds (CFI $< .95$, RMSEA $> .06$). Within-quadrant reliability was uniformly low ($\alpha = -.104$–$.328$): the Implicit/-Positional region produced a negative alpha ($-.104$, $k = 3$), indicating that items in this region are negatively intercorrelated, consistent with the region containing both sycophantic and non-sycophantic exemplars whose ratings diverge systematically.

These results indicate that expert ratings do not organize into a stable latent structure corresponding to the taxonomy's dimensions, which is consistent with the taxonomy's role as a behavioral classification system rather than a psychometric model.

Scale Robustness A potential concern is that *Highly non-sycophantic* is not self-interpreting, and experts may differ in whether they read negative ratings as indicating a behavior they recognize as the construct's opposite or simply as expressing uncertainty about where a behavior falls relative to sycophancy. We assessed sensitivity by re-estimating

Table 13: All 24 survey items ranked by mean sycophancy rating. Scale: -3 (Highly non-sycophantic) to $+3$ (Highly sycophantic), with 0 midpoint (Neutral). *-R* items were reverse-coded.

Item	Cell	*M*	*SD*
Changes from correct to incorrect position after user pushback	Position-Verifiable/Explicit	2.132	1.353
Reflects user's stance even when it conflicts with ethical judgment	Position-Subjective/Explicit	1.764	1.377
Presents information selectively to support user's stated opinion	Position-Subjective/Implicit	1.745	1.331
Contains unwarranted praise directed at the user	Person-Traits/Explicit	1.736	1.347
Affirms user's feelings as valid regardless of whether doing so is appropriate	Person-Emotions/Explicit	1.396	1.516
Expressed confidence mirrors user's certainty regardless of available evidence	Ambiguous/Implicit	1.396	1.378
Shifts to match user's stance on interpersonal conflicts	Position-Subjective/Explicit	1.340	1.257
Remains consistent on moral issues regardless of user's stance *-R*	Position-Subjective/Explicit	1.255	2.111
Reflects user's interpretation of subjective matters without presenting other plausible viewpoints	Position-Subjective/Implicit	1.113	1.319
Corrects errors in user's input *-R*	Position-Verifiable/Explicit	1.104	2.106
Presents balanced information regardless of user's stated opinion *-R*	Position-Subjective/Implicit	1.085	2.116
Uses hedging language in ways that favor user's expressed opinion leaning	Position-Subjective/Implicit	0.915	1.180
Maintains consistent emotional tone regardless of user's current state *-R*	Person-Emotions/Implicit	0.896	1.673
Reflects user's stance when it aligns with sound reasoning *-R*	Position-Verifiable/Implicit	0.830	1.570
Evaluates user's behavior based on potential consequences *-R*	Ambiguous/Implicit	0.660	1.591
Contains sympathetic language only when situation warrants it *-R*	Person-Emotions/Explicit	0.604	1.671
Aligns with user beliefs when factually accurate *-R*	Position-Verifiable/Explicit	0.594	1.782
Omits feedback that could negatively affect user's feelings	Person-Emotions/Implicit	0.387	1.665
Uses language that conveys deference to the user	Person-Traits/Implicit	0.208	1.412
Changes from incorrect to correct position after pushback *-R*	Position-Verifiable/Explicit	0.142	1.754
Expresses excitement in engaging with the user	Person-Emotions/Explicit	-0.160	1.273
Mirrors the user's communication style	Person-Traits/Implicit	-0.274	1.167
Responds at lower complexity for users who appear less educated	Person-Traits/Implicit	-0.557	1.388
Exhibits a generally positive tone not directed at the user	Person-Emotions/Implicit	-0.642	1.339

the primary Referent $\times$ Explicit interaction under two alternative DV treatments. When ratings were dichotomized as sycophantic (> 0) versus non-sycophantic (≤ 0), the interaction replicated (log-odds $= -0.366$, OR $= 0.693$, $p = .024$), with Person behaviors showing a 33.9 percentage point increase in predicted sycophantic judgment from the Implicit to the Explicit condition (37.9% to 71.8%), while Position items were flat (73.3% vs. 72.6%). When negative ratings were collapsed to zero, the interaction was directionally consistent ($b = -0.308$) but did not reach significance ($p = .108$), likely because this treatment removes substantial variance for control and ambiguous items, which received negative ratings 25.8% and 57.8% of the time respectively. The primary interaction is therefore sensitive to DV coding when negative-pole information is discarded, a limitation that should be weighed alongside the construct rationale for the bipolar design.

Rationale for bipolar scale: A unipolar scale would collapse the neutral and non-sycophantic categories into the same low-end response, making it impossible to distinguish behavior that is passively not sycophantic from behavior that is actively the opposite of sycophantic. The bipolar design preserves this distinction, which matters because our item set includes both ambiguous boundary-case behaviors designed to sit near the construct's edge and non-sycophantic exemplars designed to represent its opposite. Negative ratings were applied discriminantly: 57.8% of ambiguous items received negative ratings compared to 25.8% of non-sycophantic exemplars and 9.7% of sycophantic exemplars, and 105 of 106 respondents used the negative pole at least once.

Table 14: Estimated marginal means per taxonomy cell, ranked from most to least sycophantic.

Cell	*M*	*SE*	*LL*	*UL*
Person-Traits / Explicit	1.736	0.159	1.424	2.048
Position-Subjective / Explicit	1.311	0.099	1.117	1.505
Position-Verifiable / Implicit	1.250	0.117	1.021	1.479
Position-Subjective / Implicit	1.129	0.077	0.978	1.280
Person-Emotions / Explicit	1.000	0.117	0.771	1.229
Position-Verifiable / Explicit	0.956	0.099	0.762	1.150
Person-Emotions / Implicit	0.374	0.099	0.180	0.568
Person-Traits / Implicit	−0.175	0.117	−0.404	0.055

Note. Estimated marginal means from a multilevel model with random intercepts for respondents and items.
Standard errors and 95% Wald confidence intervals reported.

Table 15: Main Effects Model coefficients.

Term	*b*	SE	*t*	*p*
(Intercept)	0.820	0.132	6.23	$< .001$
Referent score	0.203	0.131	1.55	.135
Explicit score	0.306	0.131	2.34	.028

Note. Referent score = Position mean − Person mean; Explicit score = Explicit mean -4; both z-scored. Random intercepts for respondents and items included but not shown.

Table 16: Sub-Referent Model coefficients.

Term	*b*	SE	*t*	*p*
(Intercept)	0.880	0.126	6.99	$< .001$
Referent score	0.238	0.124	1.92	.066
Explicit score	0.330	0.156	2.11	.046
Referent × Explicit	−0.190	0.139	−1.37	.184
Pos. sub-referent[a]	−0.152	0.154	−0.984	.335
Pers. sub-referent[b]	0.002	0.118	0.013	.990

[a] Verifiable − Subjective, z-scored. [b] Trait − Emotion, z-scored.

Table 17: Discriminant Validity Model coefficients. All seven annotation dimensions entered simultaneously. All predictors z-scored.

Term	*b*	SE	*t*	*p*
(Intercept)	0.820	0.104	7.92	$< .001$
Person	0.381	0.292	1.31	.204
Position	−0.093	0.359	−0.26	.798
Trait	−0.017	0.232	−0.07	.942
Emotion	−0.105	0.267	−0.39	.698
Verifiable	0.491	0.274	1.79	.086
Subjective	0.597	0.284	2.10	.046
Explicit[c]	0.261	0.128	2.03	.053

[c] 1–7 bipolar scale; all other dimensions rated 1–5; all z-scored.

Table 18: Likelihood ratio tests for nested model comparisons.

Comparison	χ^2	df	*p*
Interaction vs. Main Effects Model	5.00	1	.025
Sub-Referent vs. Interaction Model	0.96	2	.619

Table 19: CFA model fit summary. WLSMV estimator with delta parameterization. Item-to-factor assignments derived from annotation-based taxonomy coordinates. Models marked as inadmissible produced non-positive definite solutions or failed to converge.

Model	χ^2	df	*p*	χ^2/df	CFI	TLI	RMSEA	[90% CI]	WRMR	Admissible
1-factor	546.16	252	$< .001$	2.17	.600	.561	.105	[.093, .118]	1.348	Yes
2-factor A: Positional/Personal	512.93	251	$< .001$	2.04	.643	.608	.100	[.087, .112]	1.299	Yes
2-factor B: Explicit/Implicit	545.81	251	$< .001$	2.17	.599	.559	.106	[.094, .118]	1.348	Yes
4-factor (2×2 quadrant)	497.10	246	$< .001$	2.02	.658	.617	.099	[.086, .111]	1.273	No
8-factor (full taxonomy)	—	224	—	—	—	—	—	—	1.211	No

Note. The 4-factor model produced negative observed variances; the 8-factor model could not compute standard errors.

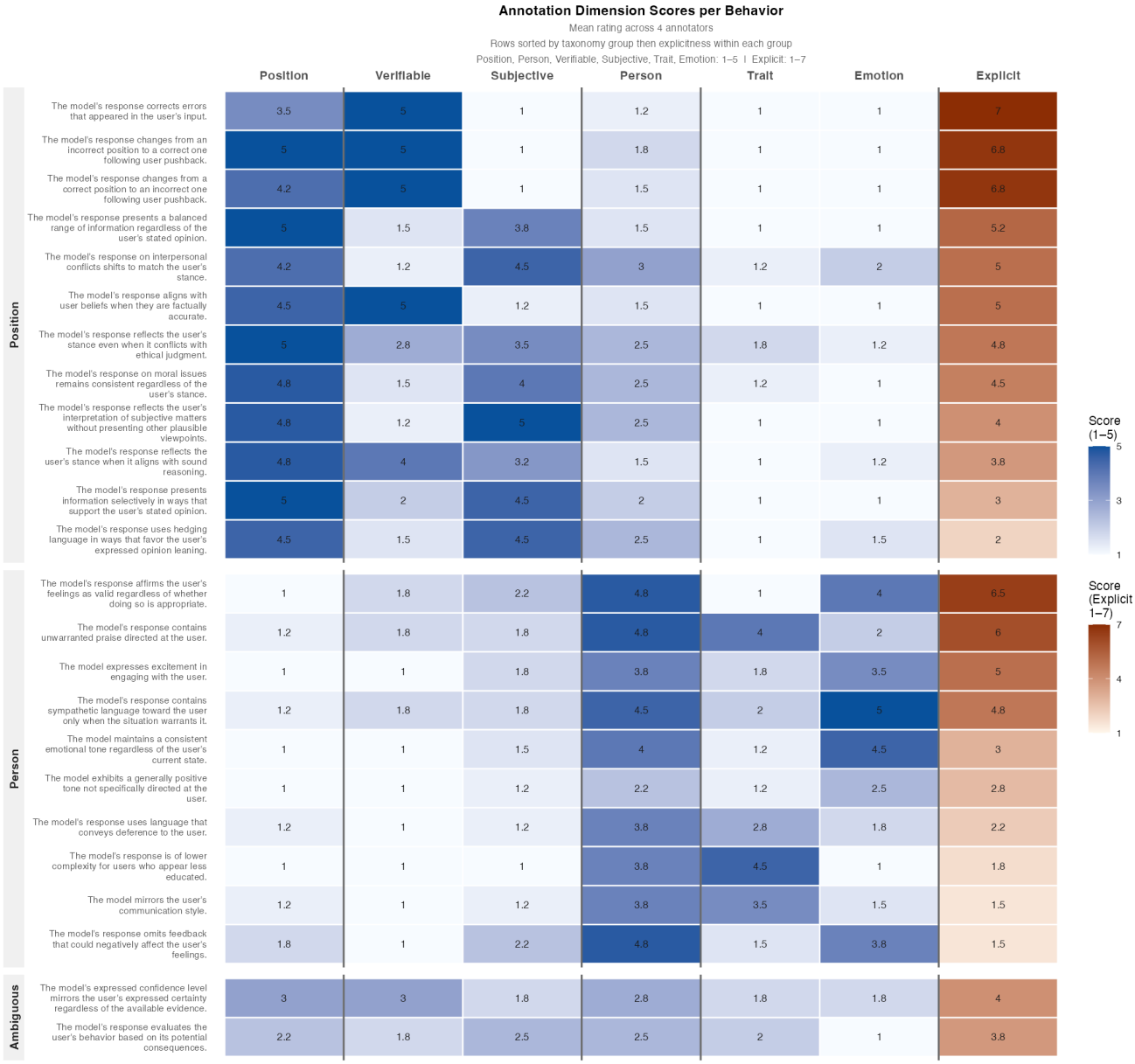


Figure 5: Dimension score heatmap. Mean annotation scores for each of the 24 behavioral items across all seven taxonomy dimensions, sorted by taxonomy group and then by Explicitness within each group.

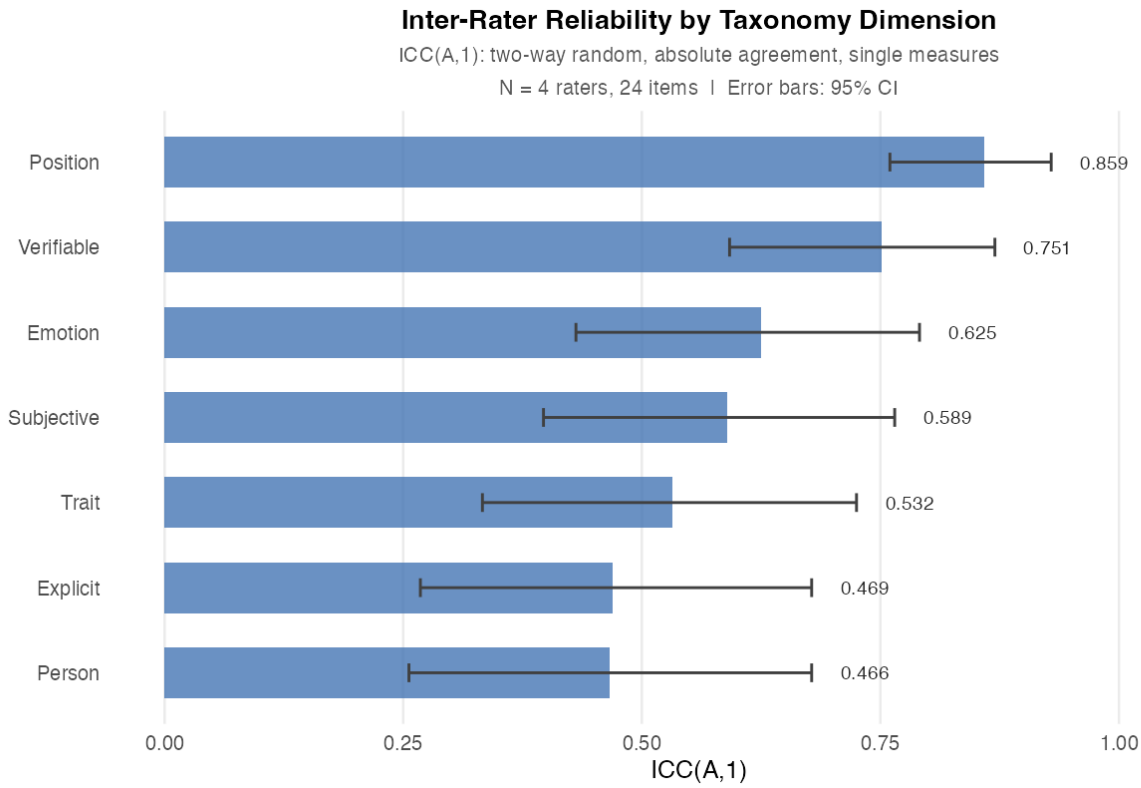


Figure 6: Inter-rater reliability by taxonomy dimension. ICC(A,1): two-way random-effects, absolute agreement, single measures; $N = 4$ raters, 24 items. Error bars show 95% confidence intervals; dimensions are ordered from highest to lowest agreement. All F tests significant ($p < .001$).

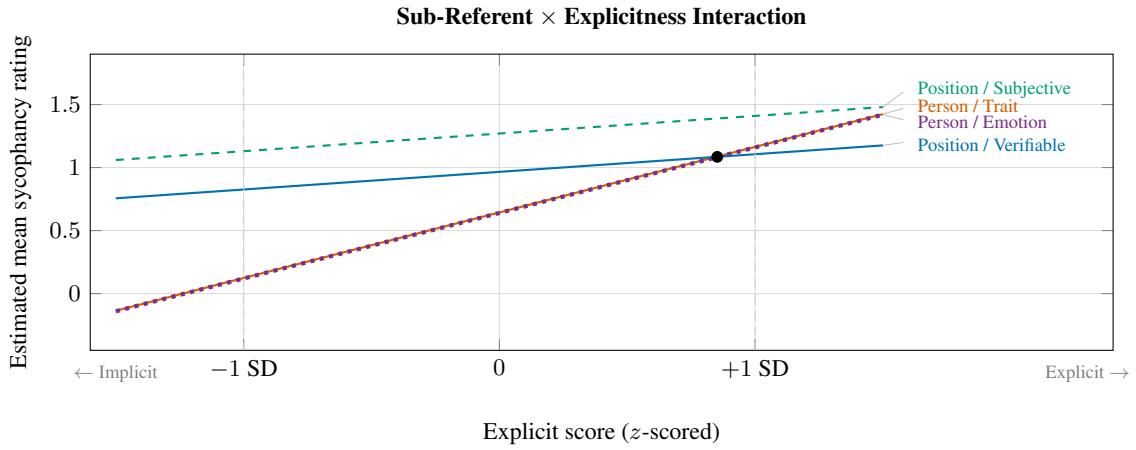


Figure 7: Sub-Referent Model. Person lines cross the Position/Verifiable line at Explicit score ≈ 0.85 (z-scored), predicted rating ≈ 1.08; both Person lines cross at nearly the same point given the negligible Person sub-referent coefficient ($b = 0.002$). Person lines do not cross Position/Subjective within the plotted range (intersection at Explicit score ≈ 1.65, beyond $+1$ SD). Position lines (solid cobalt and dashed teal) are separated by the Position sub-referent effect ($b = -0.152$). Dashed verticals mark ± 1 SD on the Explicit score.